

SEARCH EXAMPLE

INTRODUCTION TO ARTIFICIAL INTELLIGENT (AI)



REV: 25.03.2024

Example: The puzzle's goal is to reach the red square from the upper-left square by moving horizontally or vertically as much as the number in the square you are in.

Search tree and graph solution by using

- Uniform cost search
- Greedy search
- A* search

2	3	1	1
1	2	3	2
4	1	4	2
2	2	3	

FRONTIER

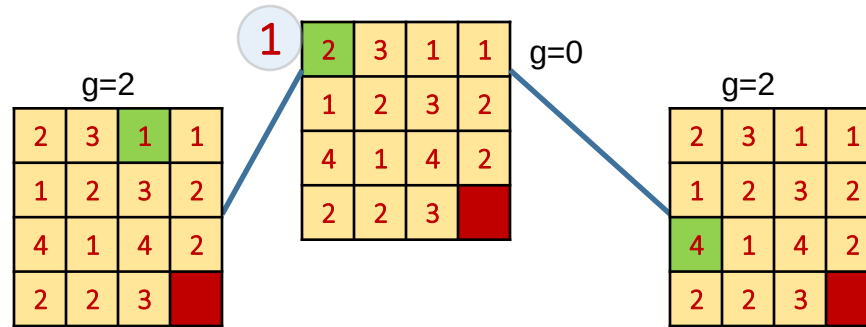
$[(1,1)_0]$

Ex 1: Uniform Cost Search

2	3	1	1
1	2	3	2
4	1	4	2
2	2	3	

g=0

Ex 1: Uniform Cost Search

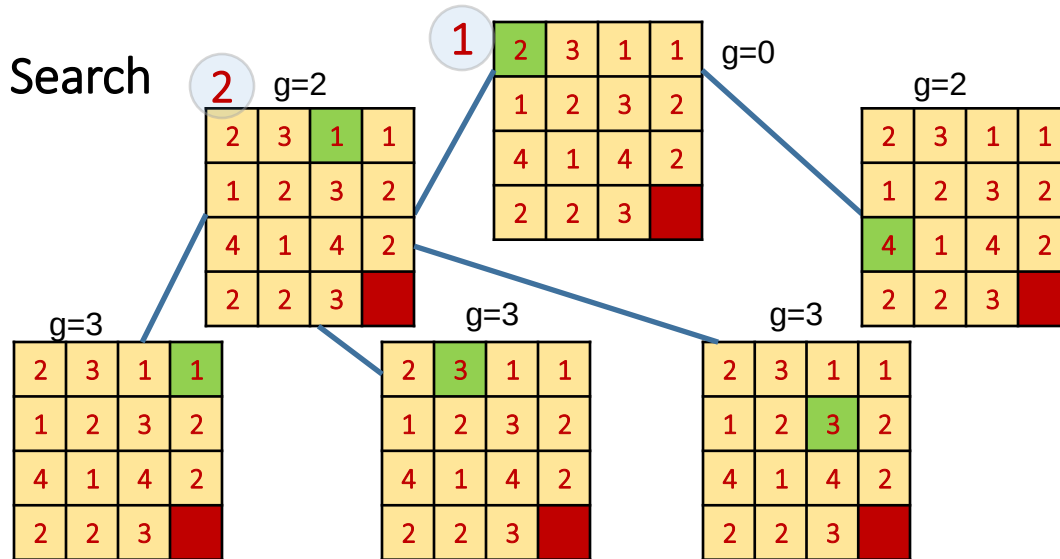


FRONTIER

$[(1,1)_0]$

$[(1,3)_2, (3,1)_2]$

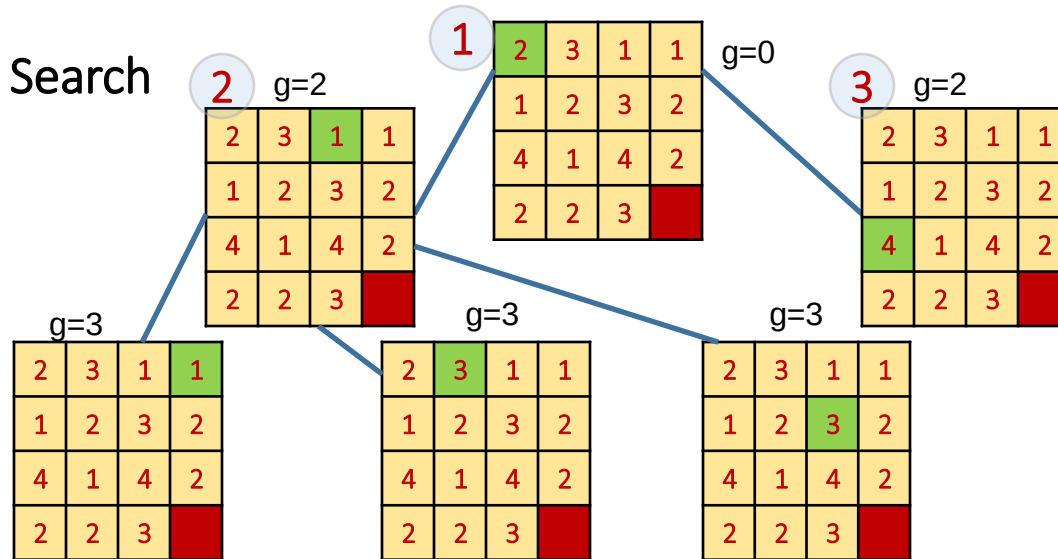
Ex 1: Uniform Cost Search



FRONTIER

$[(1,1)_0]$
 $[(1,3)_2, (3,1)_2]$
 $[(3,1)_2, (1,4)_3, (1,2)_3, (2,3)_3]$

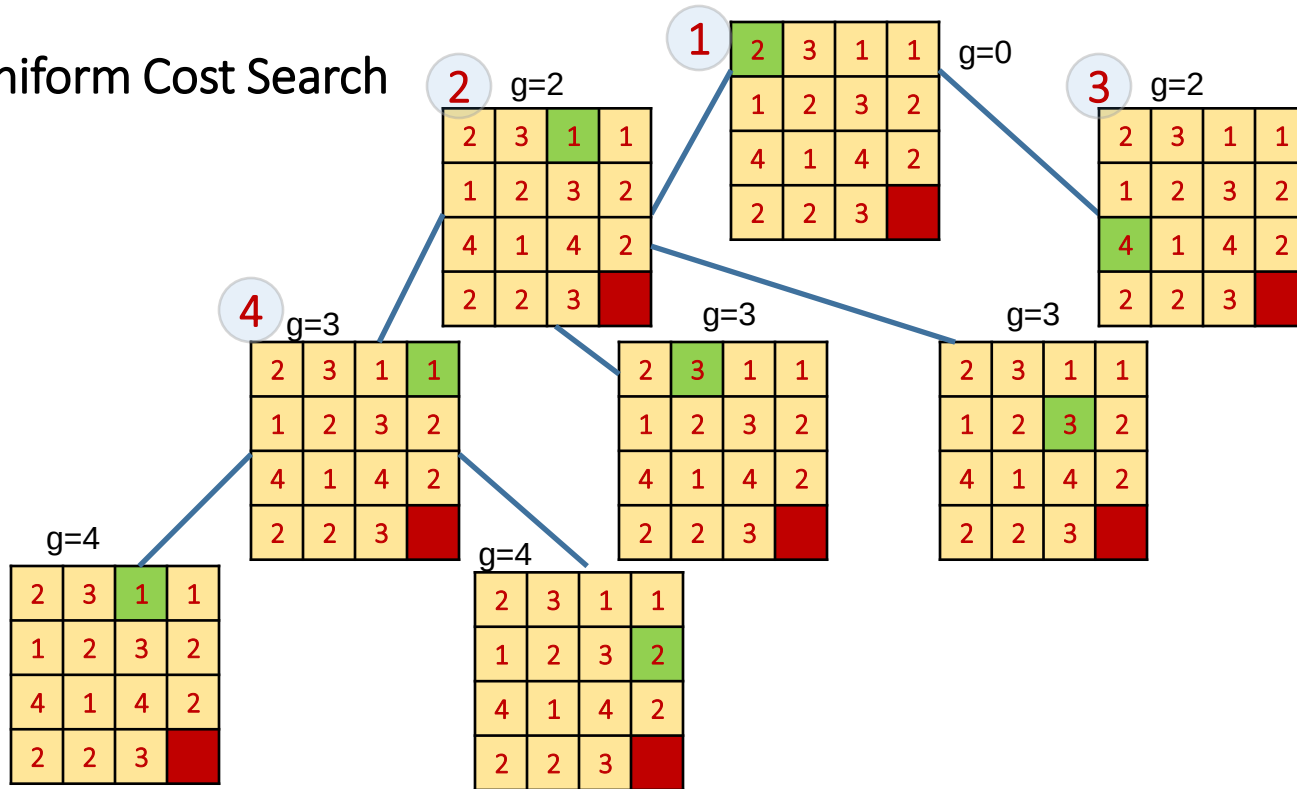
Ex 1: Uniform Cost Search



FRONTIER

$[(1,1)_0]$
 $[(1,3)_2, (3,1)_2]$
 $[(3,1)_2, (1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,4)_3, (1,2)_3, (2,3)_3]$

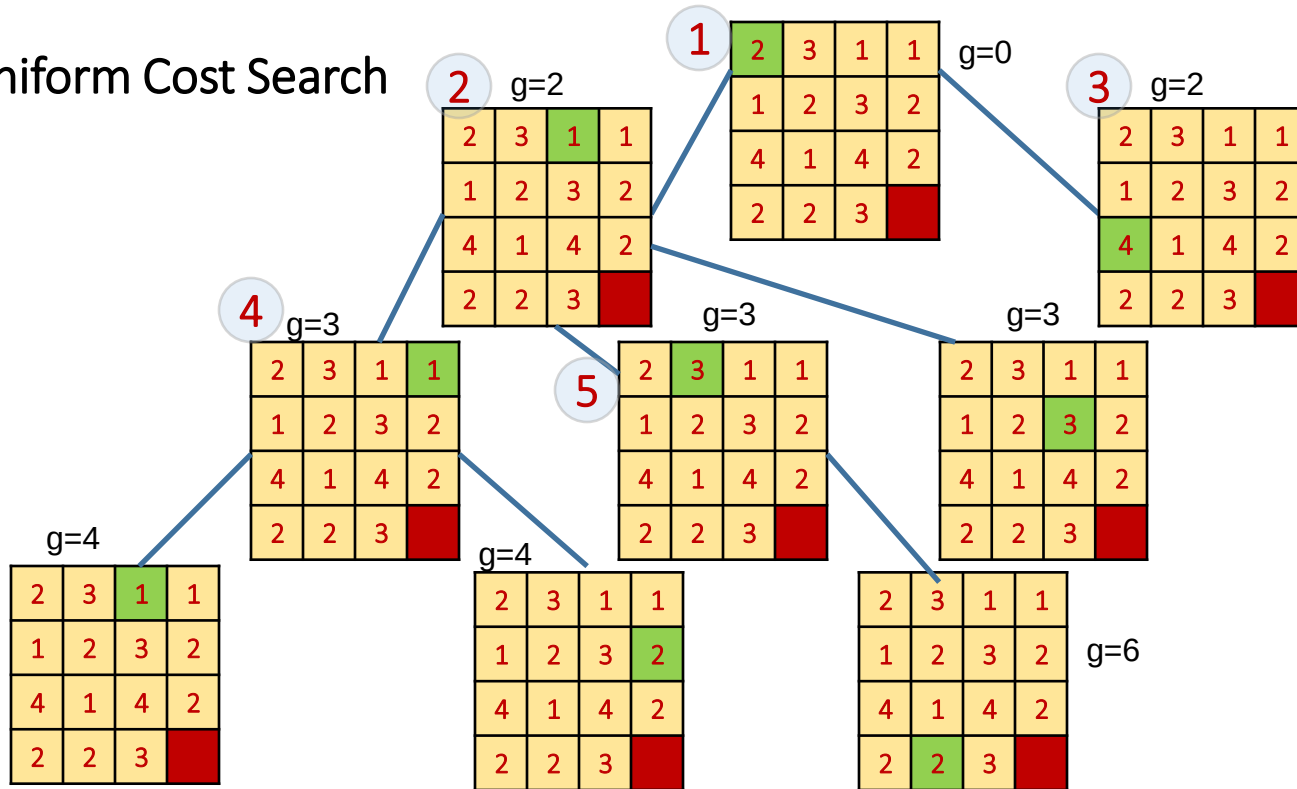
Ex 1: Uniform Cost Search



FRONTIER

$[(1,1)_0]$
 $[(1,3)_2, (3,1)_2]$
 $[(3,1)_2, (1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,2)_3, (2,3)_3, (1,3)_4, (2,4)_4]$

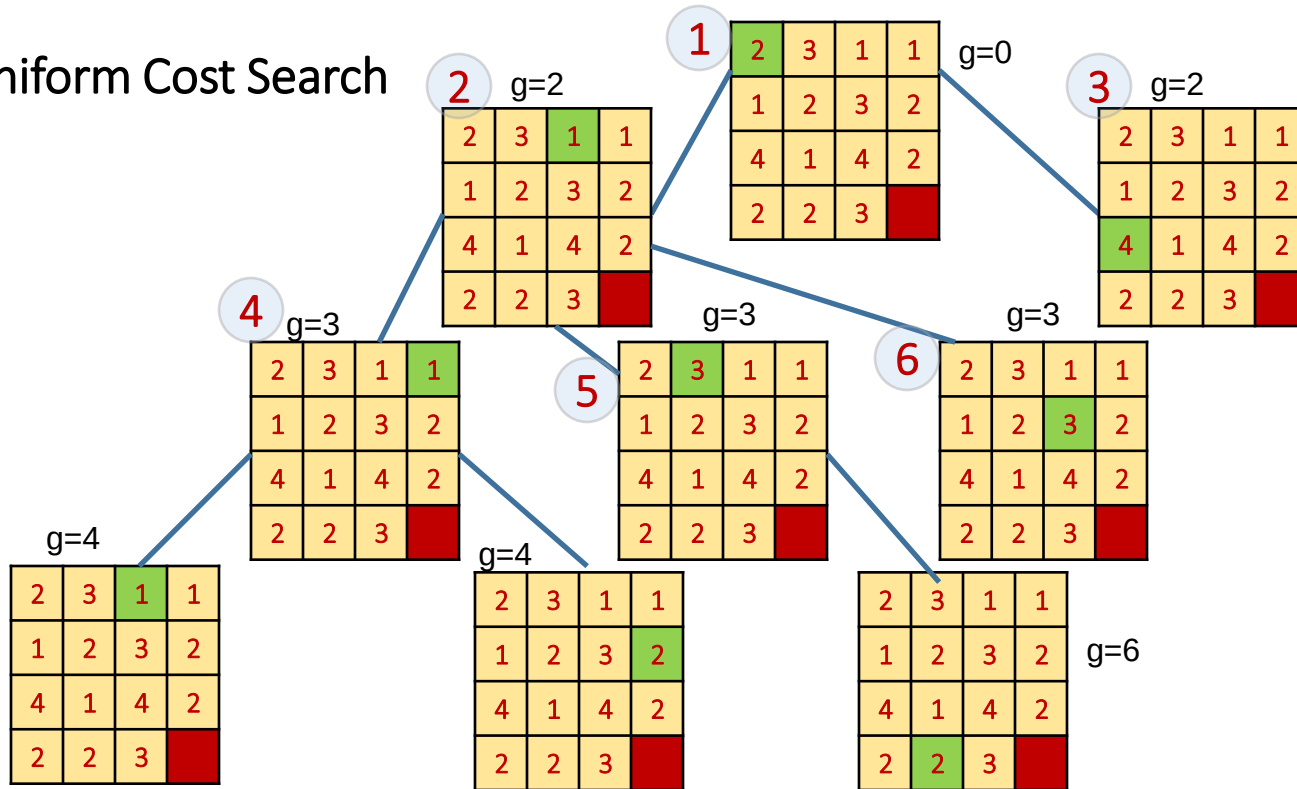
Ex 1: Uniform Cost Search



FRONTIER

$[(1,1)_0]$
 $[(1,3)_2, (3,1)_2]$
 $[(3,1)_2, (1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,2)_3, (2,3)_3, (1,3)_4, (2,4)_4]$
 $[(2,3)_3, (1,3)_4, (2,4)_4, (4,2)_6]$

Ex 1: Uniform Cost Search



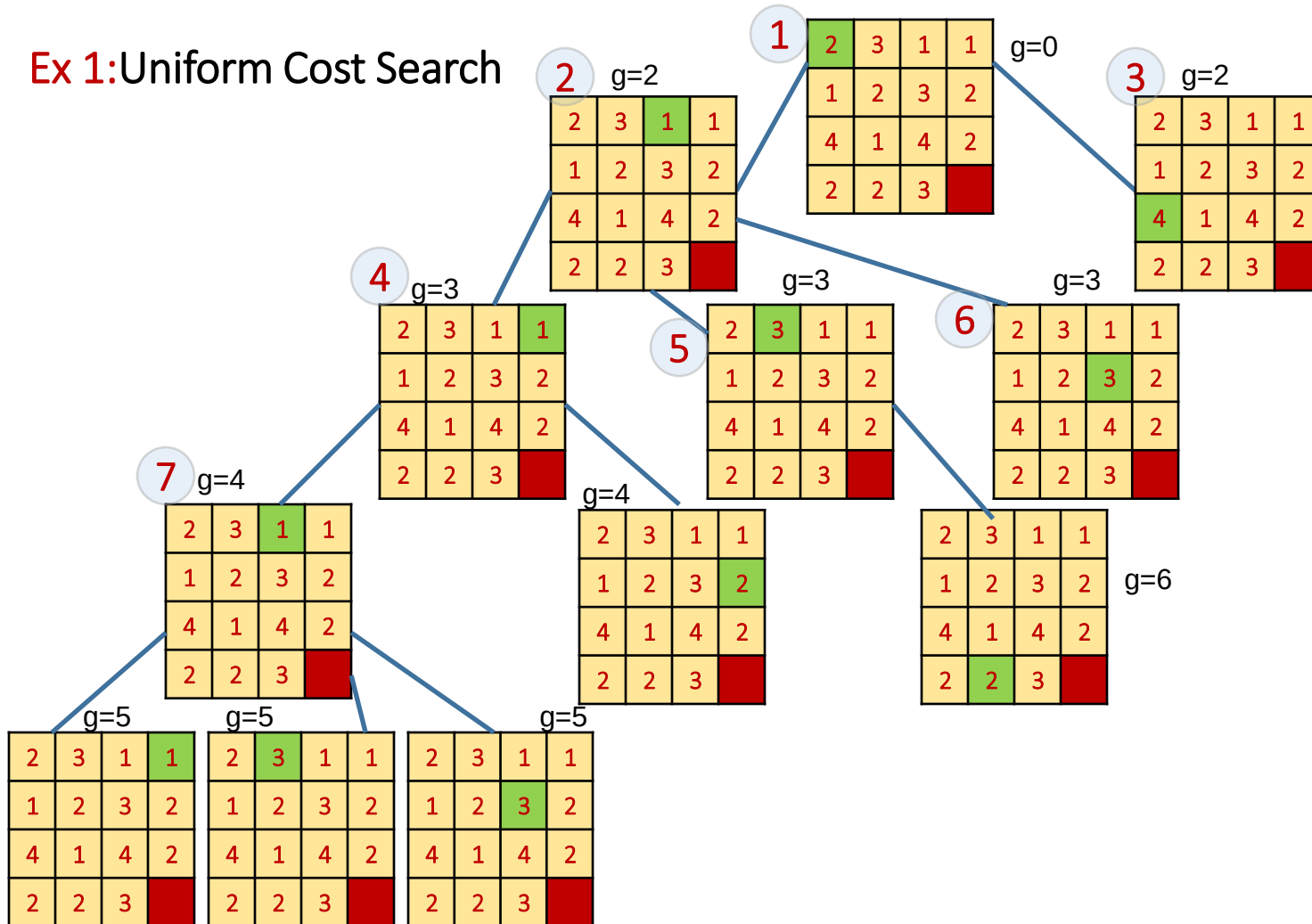
FRONTIER

$[(1,1)_0]$
 $[(1,3)_2, (3,1)_2]$
 $[(3,1)_2, (1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,2)_3, (2,3)_3, (1,3)_4, (2,4)_4]$
 $[(2,3)_3, (1,3)_4, (2,4)_4, (4,2)_6]$
 $[(1,3)_4, (2,4)_4, (4,2)_6]$

Ex 1: Uniform Cost Search

FRONTIER

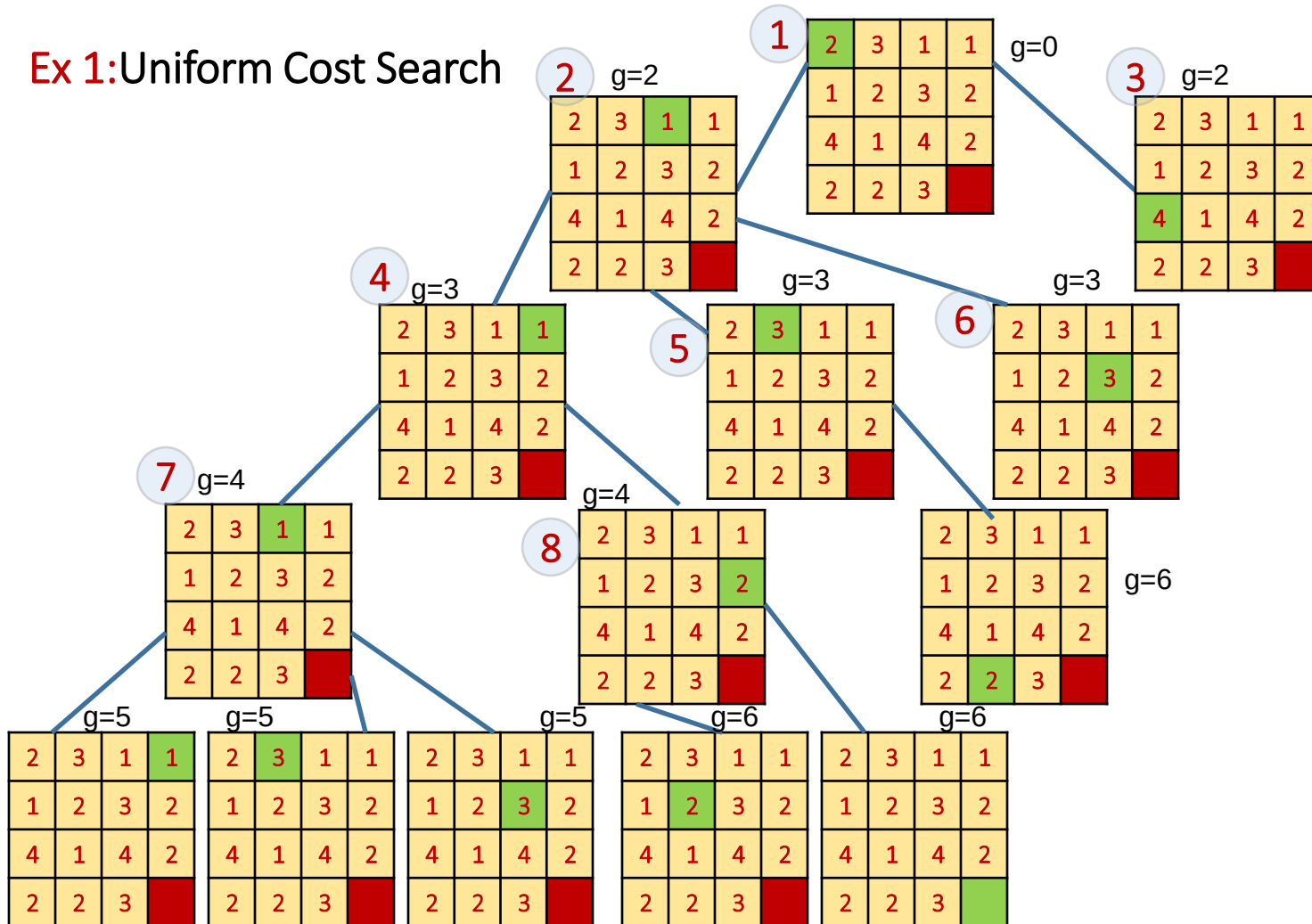
$[(1,1)_0]$
 $[(1,3)_2, (3,1)_2]$
 $[(3,1)_2, (1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,2)_3, (2,3)_3, (1,3)_4, (2,4)_4]$
 $[(2,3)_3, (1,3)_4, (2,4)_4, (4,2)_6]$
 $[(1,3)_4, (2,4)_4, (4,2)_6]$
 $[(2,4)_4, (1,4)_5, (1,2)_5, (2,3)_5, (4,2)_6]$



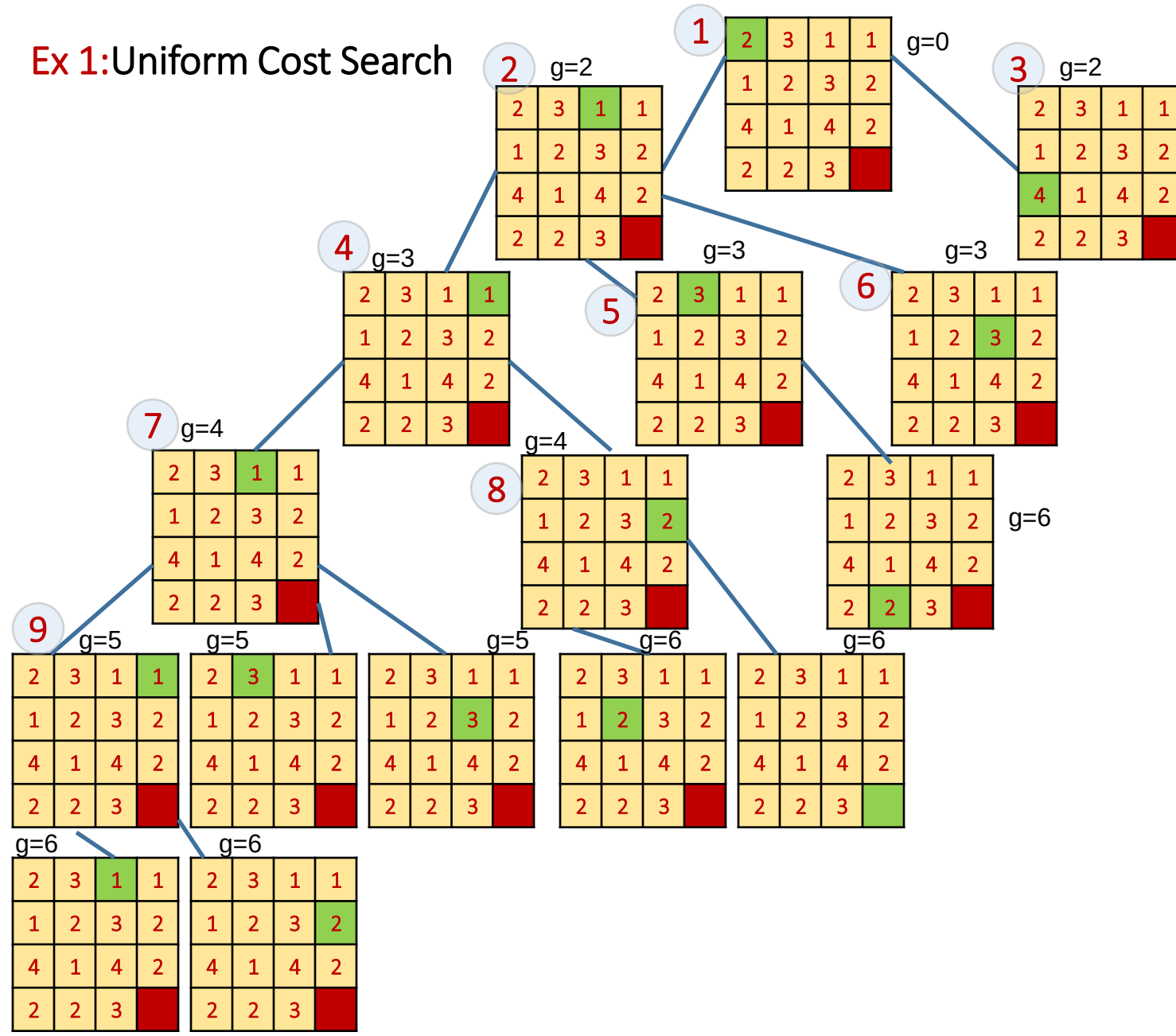
Ex 1: Uniform Cost Search

FRONTIER

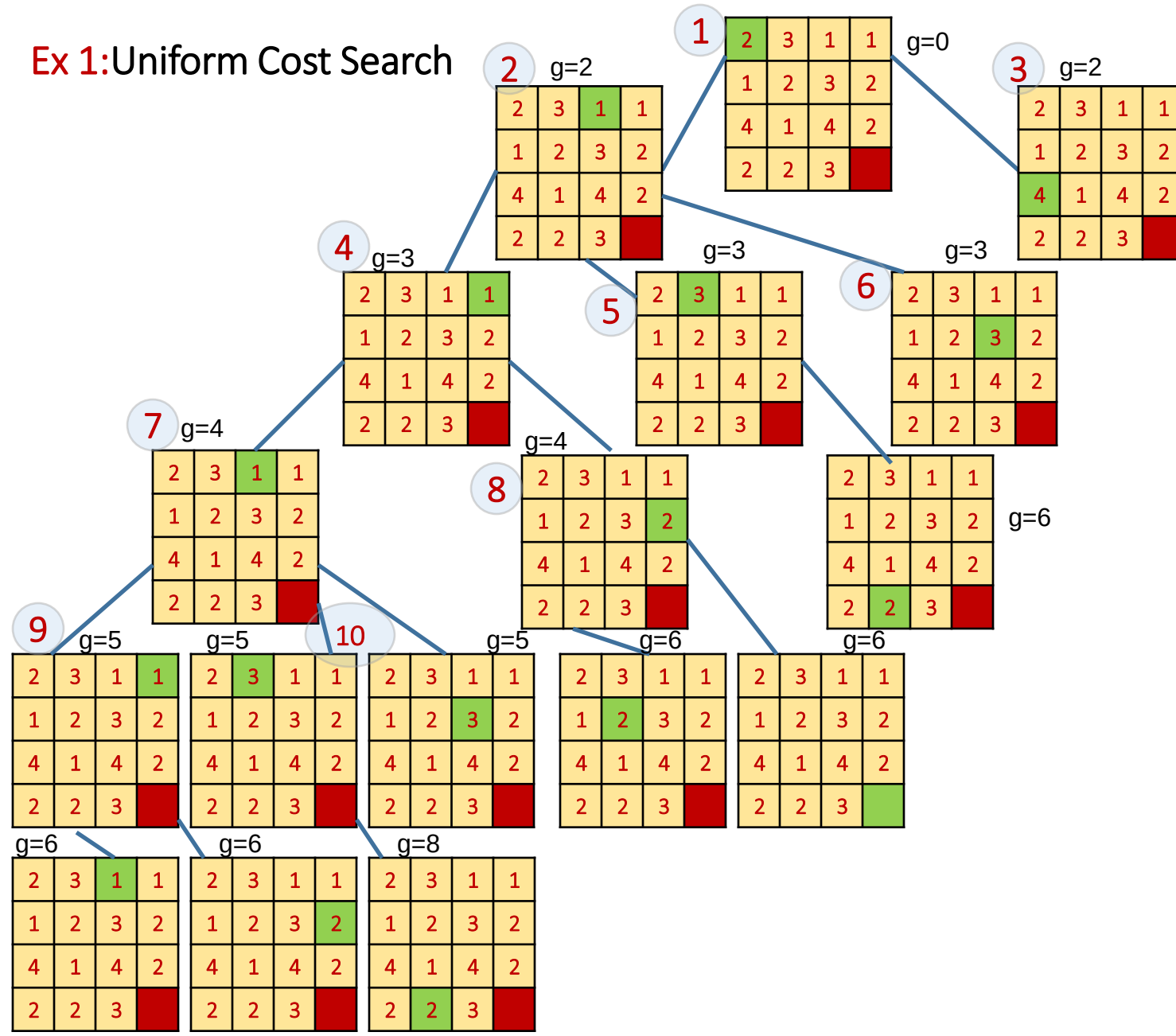
$[(1,1)_0]$
 $[(1,3)_2, (3,1)_2]$
 $[(3,1)_2, (1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,2)_3, (2,3)_3, (1,3)_4, (2,4)_4]$
 $[(2,3)_3, (1,3)_4, (2,4)_4, (4,2)_6]$
 $[(1,3)_4, (2,4)_4, (4,2)_6]$
 $[(2,4)_4, (1,4)_5, (1,2)_5, (2,3)_5, (4,2)_6]$
 $[(1,4)_5, (1,2)_5, (2,3)_5, (4,2)_6, (2,2)_6, (4,4)_6]$



FRONTIER


$$\begin{aligned} & [(1,1)_0] \\ & [(1,3)_2, (3,1)_2] \\ & [(3,1)_2, (1,4)_3, (1,2)_3, \\ & (2,3)_3] \\ & [(1,4)_3, (1,2)_3, (2,3)_3] \\ & [(1,2)_3, (2,3)_3, (1,3)_4, \\ & (2,4)_4] \\ & [(2,3)_3, (1,3)_4, (2,4)_4, \\ & (4,2)_6] \\ & [(1,3)_4, (2,4)_4, (4,2)_6] \\ & [(2,4)_4, (1,4)_5, (1,2)_5, \\ & (2,3)_5, (4,2)_6] \\ & [(1,4)_5, (1,2)_5, (2,3)_5, \\ & (4,2)_6, (2,2)_6, (4,4)_6] \\ & [(1,2)_5, (2,3)_5, (4,2)_6, \\ & (2,2)_6, (4,4)_6, (1,3)_6, \\ & (2,4)_6] \end{aligned}$$

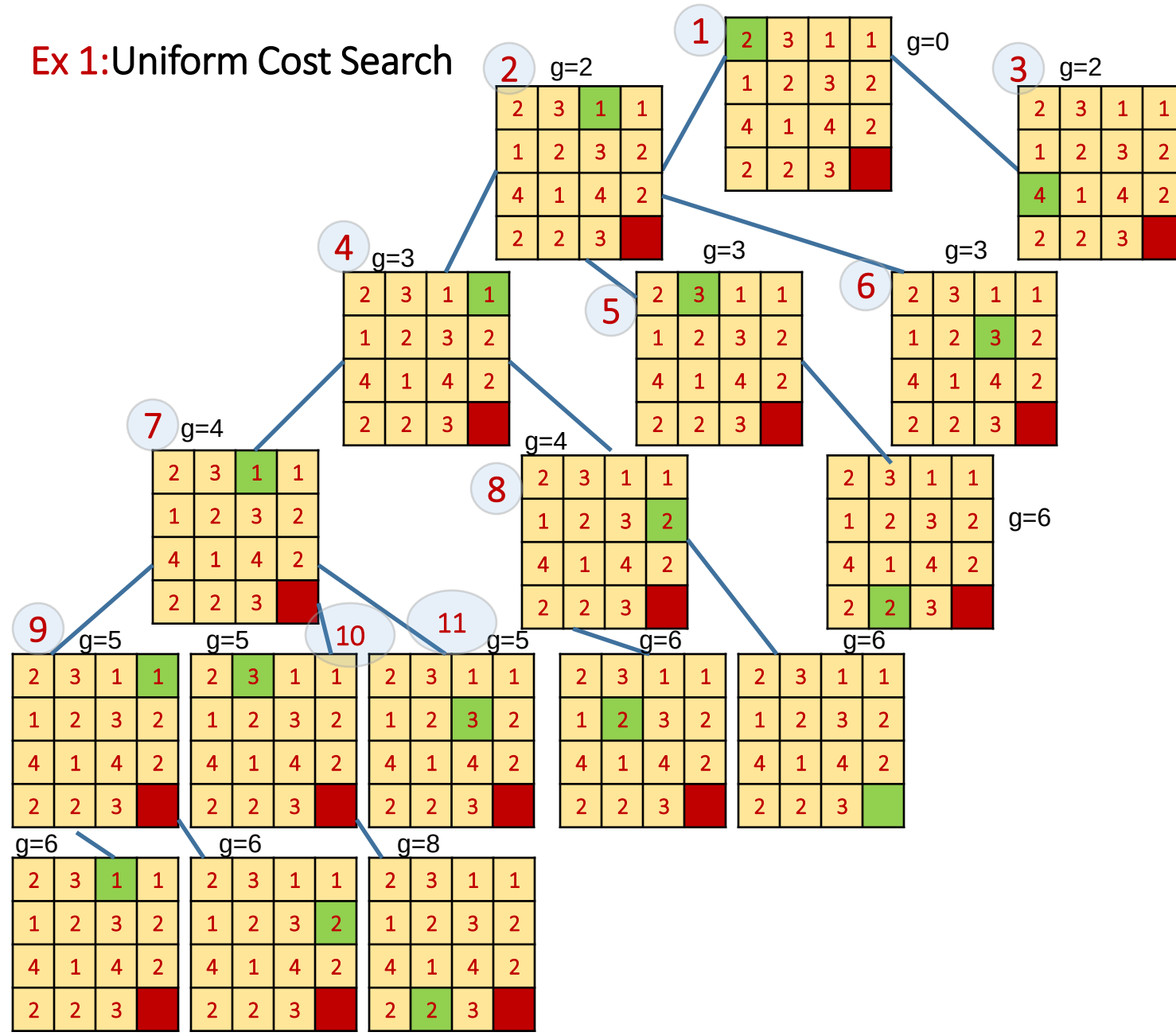
Ex 1: Uniform Cost Search



FRONTIER

$[(1,1)_0]$
 $[(1,3)_2, (3,1)_2]$
 $[(3,1)_2, (1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,2)_3, (2,3)_3, (1,3)_4, (2,4)_4]$
 $[(2,3)_3, (1,3)_4, (2,4)_4, (4,2)_6]$
 $[(1,3)_4, (2,4)_4, (4,2)_6]$
 $[(2,4)_4, (1,4)_5, (1,2)_5, (2,3)_5, (4,2)_6]$
 $[(1,4)_5, (1,2)_5, (2,3)_5, (4,2)_6, (2,2)_6, (4,4)_6]$
 $[(1,2)_5, (2,3)_5, (4,2)_6, (2,2)_6, (4,4)_6, (1,3)_6, (2,4)_6]$
 $[(2,3)_5, (4,2)_6, (2,2)_6, (4,4)_6, (1,3)_6, (2,4)_6, (2,4)_8]$

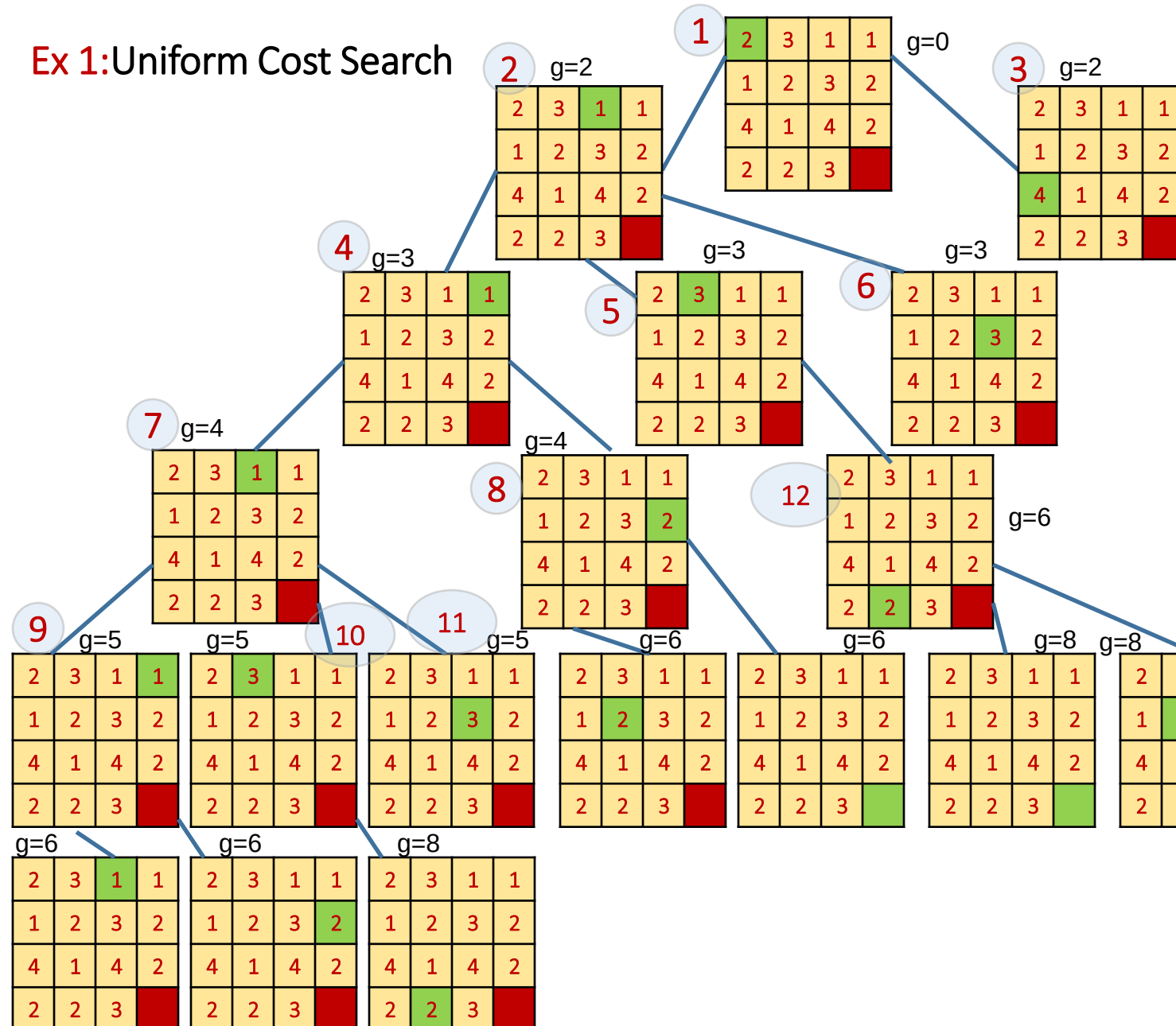
Ex 1: Uniform Cost Search



FRONTIER

$[(1,1)_0]$
 $[(1,3)_2, (3,1)_2]$
 $[(3,1)_2, (1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,2)_3, (2,3)_3, (1,3)_4, (2,4)_4]$
 $[(2,3)_3, (1,3)_4, (2,4)_4, (4,2)_6]$
 $[(1,3)_4, (2,4)_4, (4,2)_6]$
 $[(2,4)_4, (1,4)_5, (1,2)_5, (2,3)_5, (4,2)_6]$
 $[(1,4)_5, (1,2)_5, (2,3)_5, (4,2)_6, (2,2)_6, (4,4)_6]$
 $[(1,2)_5, (2,3)_5, (4,2)_6, (2,2)_6, (4,4)_6, (1,3)_6, (2,4)_6]$
 $[(2,3)_5, (4,2)_6, (2,2)_6, (4,4)_6, (1,3)_6, (2,4)_6, (2,4)_8]$
 $[(4,2)_6, (2,2)_6, (4,4)_6, (1,3)_6, (2,4)_6, (2,4)_8]$

Ex 1: Uniform Cost Search



FRONTIER

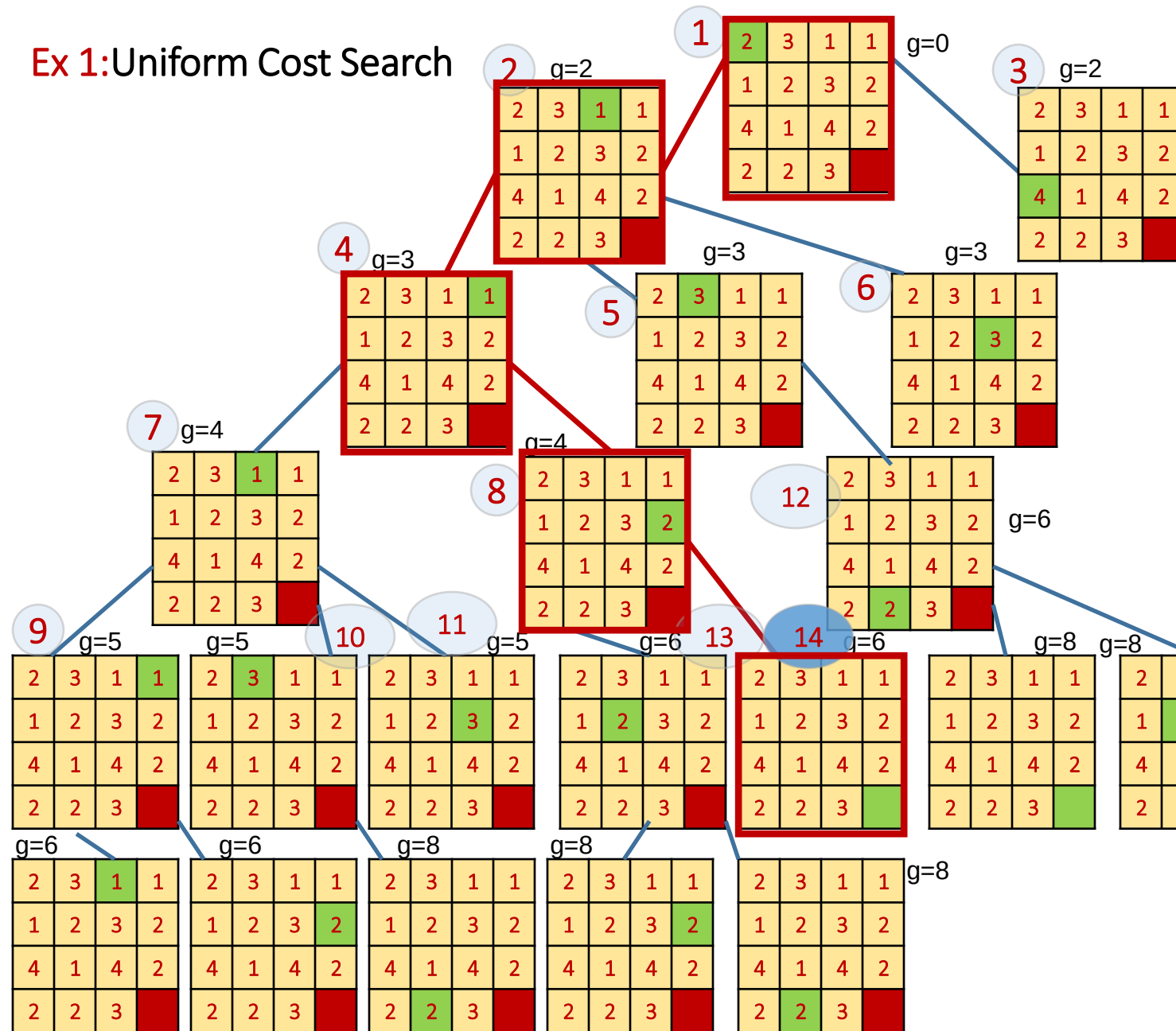
$[(1,1)_0]$
 $[(1,3)_2, (3,1)_2]$
 $[(3,1)_2, (1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,2)_3, (2,3)_3, (1,3)_4, (2,4)_4]$
 $[(2,3)_3, (1,3)_4, (2,4)_4, (4,2)_6]$
 $[(1,3)_4, (2,4)_4, (4,2)_6]$
 $[(2,4)_4, (1,4)_5, (1,2)_5, (2,3)_5, (4,2)_6]$
 $[(1,4)_5, (1,2)_5, (2,3)_5, (4,2)_6, (2,2)_6, (4,4)_6]$
 $[(1,2)_5, (2,3)_5, (4,2)_6, (2,2)_6, (4,4)_6, (1,3)_6, (2,4)_6]$
 $[(2,3)_5, (4,2)_6, (2,2)_6, (4,4)_6, (1,3)_6, (2,4)_6, (2,4)_8]$
 $[(4,2)_6, (2,2)_6, (4,4)_6, (1,3)_6, (2,4)_6, (2,4)_8]$
 $[(2,2)_6, (4,4)_6, (1,3)_6, (2,4)_6, (2,4)_8, (4,4)_8, (2,2)_8]$

$[(1,1)_0]$
 $[(1,3)_2, (3,1)_2]$
 $[(3,1)_2, (1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,2)_3, (2,3)_3, (1,3)_4, (2,4)_4]$
 $[(2,3)_3, (1,3)_4, (2,4)_4, (4,2)_6]$
 $[(1,3)_4, (2,4)_4, (4,2)_6]$
 $[(2,4)_4, (1,4)_5, (1,2)_5, (2,3)_5, (4,2)_6]$
 $[(1,4)_5, (1,2)_5, (2,3)_5, (4,2)_6, (2,2)_6, (4,4)_6]$
 $[(1,2)_5, (2,3)_5, (4,2)_6, (2,2)_6, (4,4)_6, (1,3)_6, (2,4)_6]$
 $[(2,3)_5, (4,2)_6, (2,2)_6, (4,4)_6, (1,3)_6, (2,4)_6, (2,4)_8]$
 $[(4,2)_6, (2,2)_6, (4,4)_6, (1,3)_6, (2,4)_6, (2,4)_8]$

1	1
3	2
4	2
3	

 $[(2,2)_6, (4,4)_6, (1,3)_6, (2,4)_6, (2,4)_8, (4,4)_8, (2,2)_8]$
 $[(4,4)_6, (1,3)_6, (2,4)_6, (2,4)_8, (4,4)_8, (2,2)_8, (2,4)_8, (4,2)_8]$

Ex 1: Uniform Cost Search



FRONTIER

[(1,1)₀]
 [(1,3)₂, (3,1)₂]
 [(3,1)₂, (1,4)₃, (1,2)₃, (2,3)₃]
 [(1,4)₃, (1,2)₃, (2,3)₃]
 [(1,2)₃, (2,3)₃, (1,3)₄, (2,4)₄]
 [(2,3)₃, (1,3)₄, (2,4)₄, (4,2)₆]
 [(1,3)₄, (2,4)₄, (4,2)₆]
 [(2,4)₄, (1,4)₅, (1,2)₅, (2,3)₅, (4,2)₆]
 [(1,4)₅, (1,2)₅, (2,3)₅, (4,2)₆, (2,2)₆, (4,4)₆]
 [(1,2)₅, (2,3)₅, (4,2)₆, (2,2)₆, (4,4)₆, (1,3)₆, (2,4)₆]
 [(2,3)₅, (4,2)₆, (2,2)₆, (4,4)₆, (1,3)₆, (2,4)₆, (2,4)₈]
 [(4,2)₆, (2,2)₆, (4,4)₆, (1,3)₆, (2,4)₆, (2,4)₈]
 [(2,2)₆, (4,4)₆, (1,3)₆, (2,4)₆, (2,4)₈, (4,4)₈, (2,2)₈]
 [(4,4)₆, (1,3)₆, (2,4)₆, (2,4)₈, (4,4)₈, (2,2)₈, (2,4)₈]
 [(1,3)₆, (2,4)₆, ...]

EXPLORED SET:

FRONTIER

$[(1,1)_0]$

Ex 1: Uniform Cost Search

2	3	1	1
1	2	3	2
4	1	4	2
2	2	3	

$g=0$

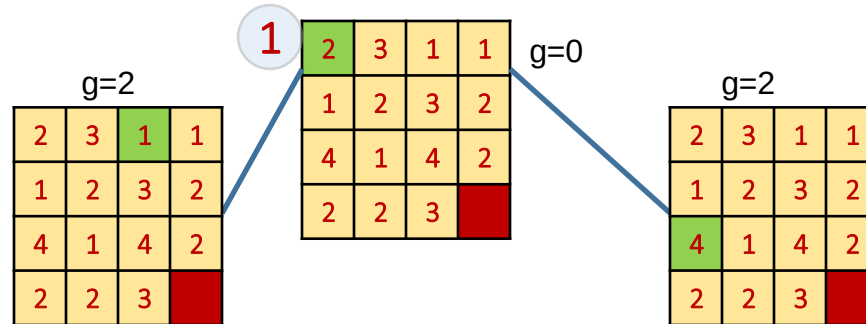
EXPLORED SET: (1,1)

FRONTIER

$[(1,1)_0]$

$[(1,3)_2, (3,1)_2]$

Ex 1: Uniform Cost Search

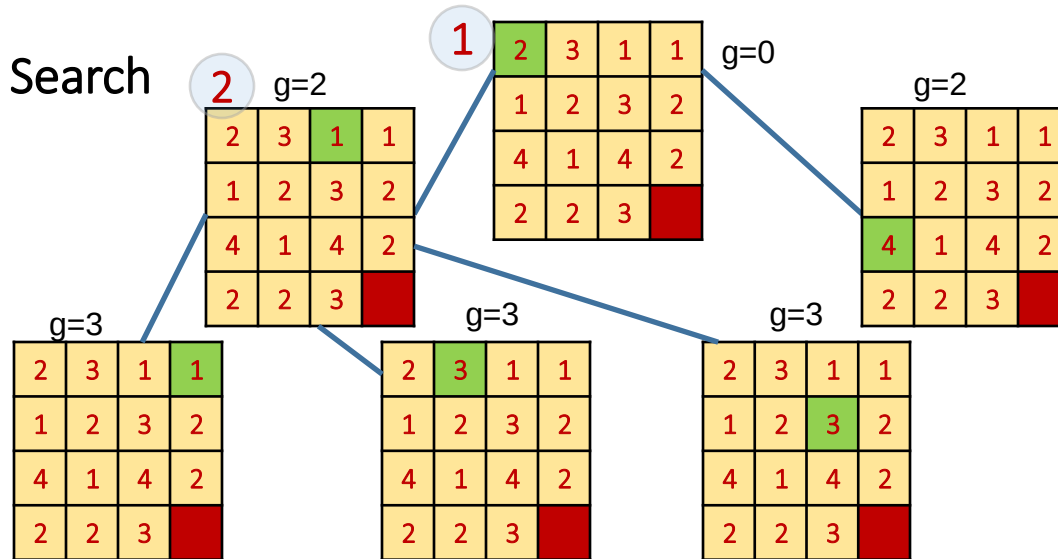


EXPLORED SET: (1,1), (1,3)

FRONTIER

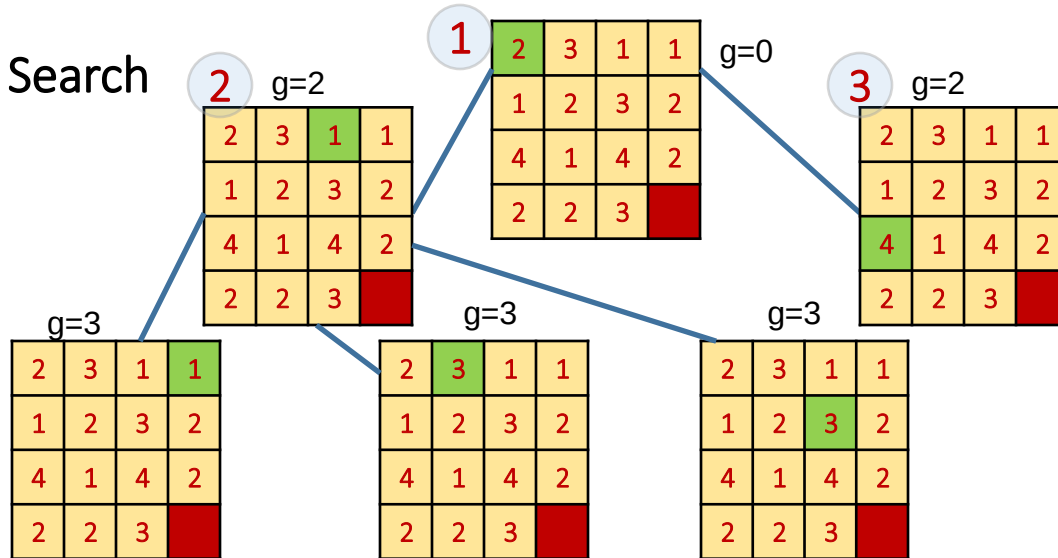
$[(1,1)_0]$
 $[(1,3)_2, (3,1)_2]$
 $[(3,1)_2, (1,4)_3, (1,2)_3, (2,3)_3]$

Ex 1: Uniform Cost Search



EXPLORED SET: (1,1), (1,3), (3,1)

Ex 1: Uniform Cost Search

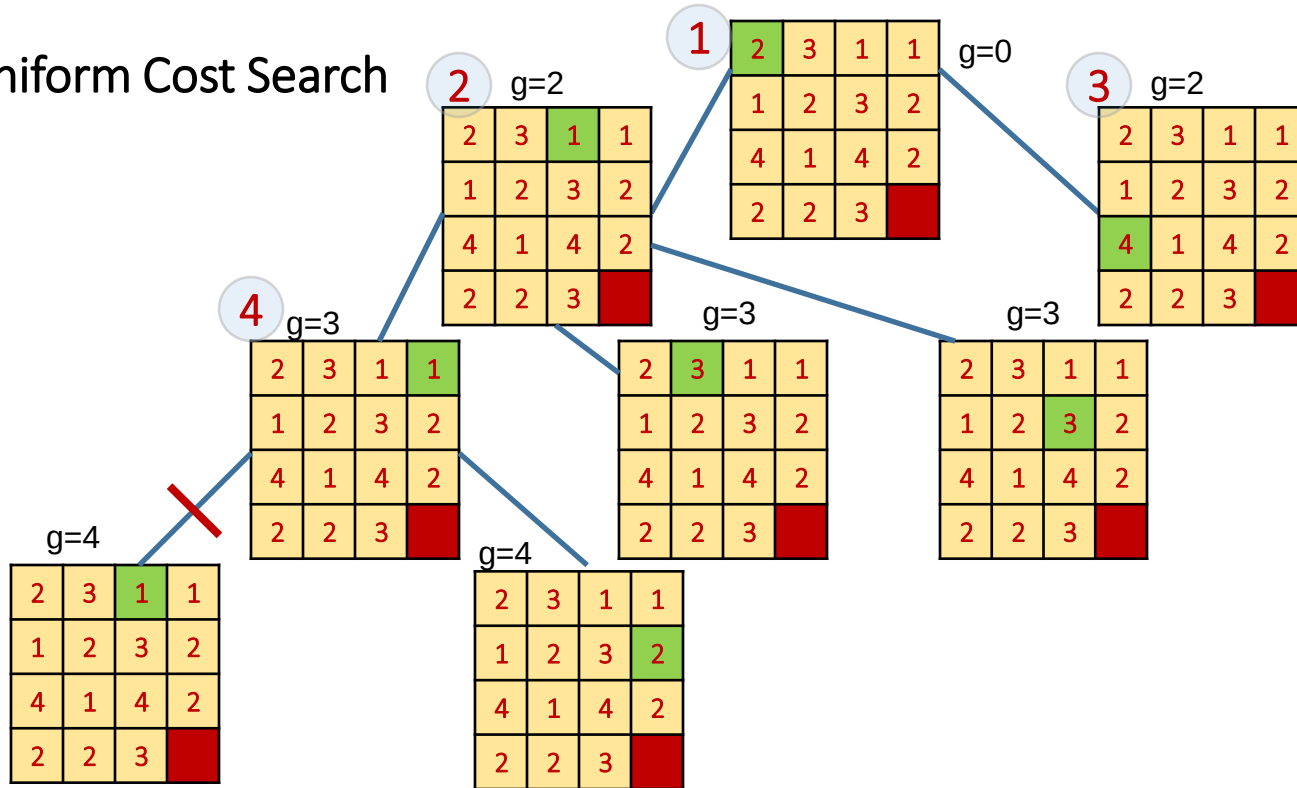


FRONTIER

$[(1,1)_0]$
 $[(1,3)_2, (3,1)_2]$
 $[(3,1)_2, (1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,4)_3, (1,2)_3, (2,3)_3]$

EXPLORED SET: (1,1), (1,3), (3,1), (1,4)

Ex 1: Uniform Cost Search



FRONTIER

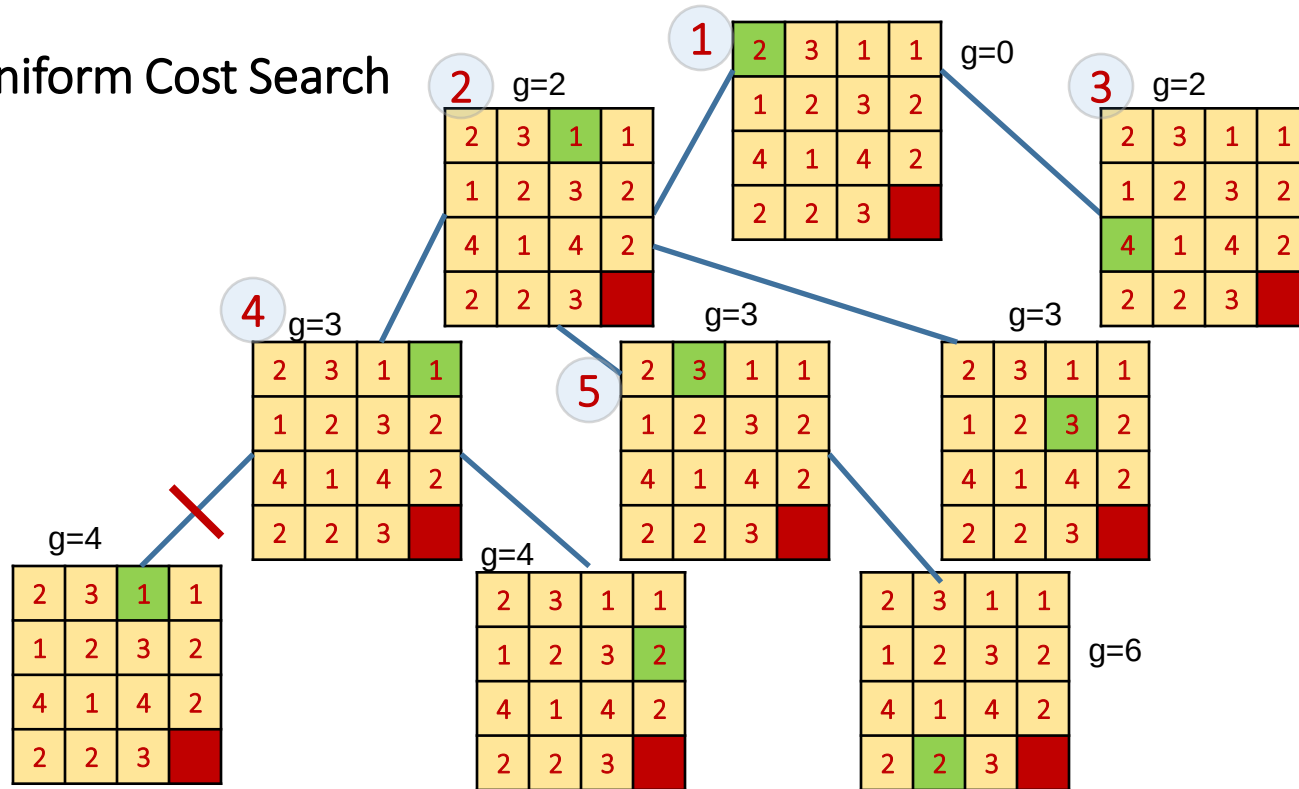
$[(1,1)_0]$
 $[(1,3)_2, (3,1)_2]$
 $[(3,1)_2, (1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,2)_3, (2,3)_3, (2,4)_4]$

EXPLORED SET: (1,1), (1,3), (3,1), (1,4), (1,2)

FRONTIER

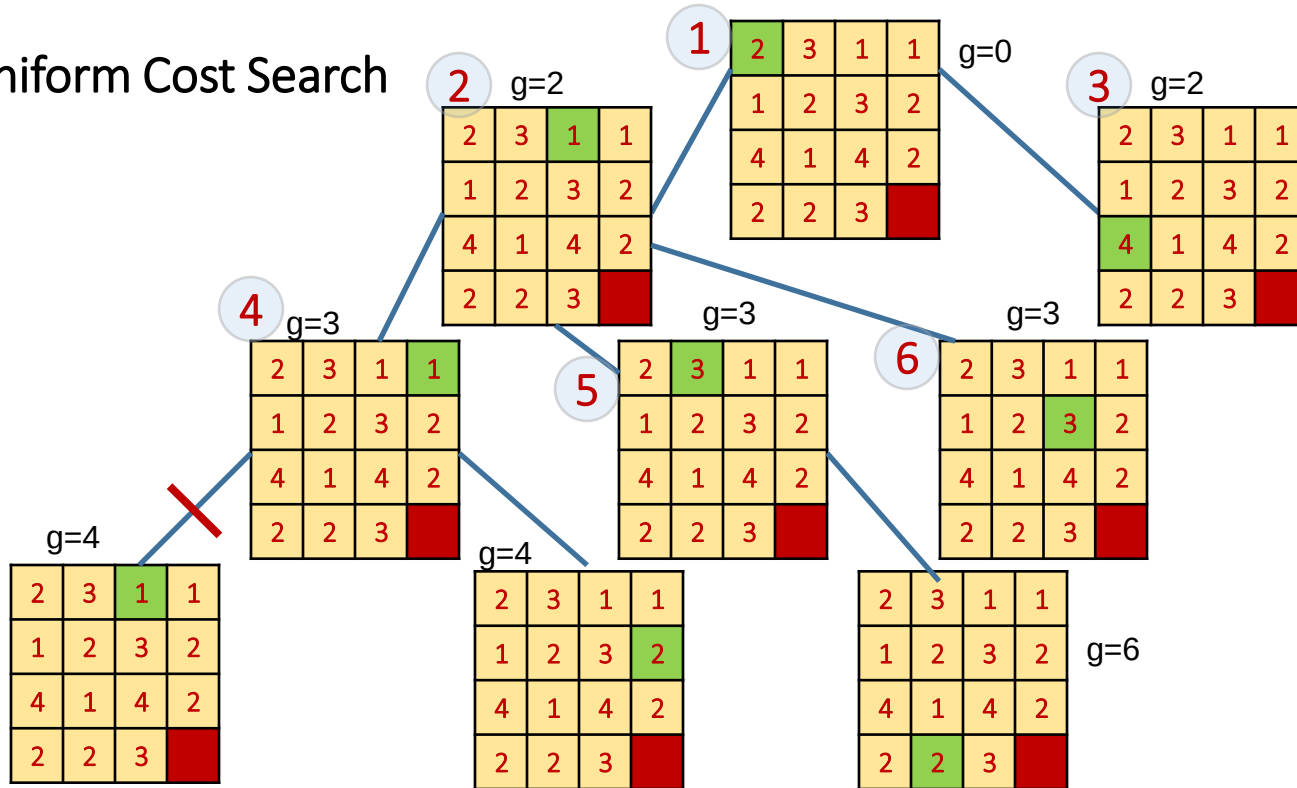
$[(1,1)_0]$
 $[(1,3)_2, (3,1)_2]$
 $[(3,1)_2, (1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,4)_3, (1,2)_3, (2,3)_3]$
 $[(1,2)_3, (2,3)_3, (2,4)_4]$
 $[(2,3)_3, (2,4)_4, (4,2)_6]$

Ex 1: Uniform Cost Search



EXPLORED SET: (1,1), (1,3), (3,1), (1,4), (1,2), (2,3)

Ex 1: Uniform Cost Search

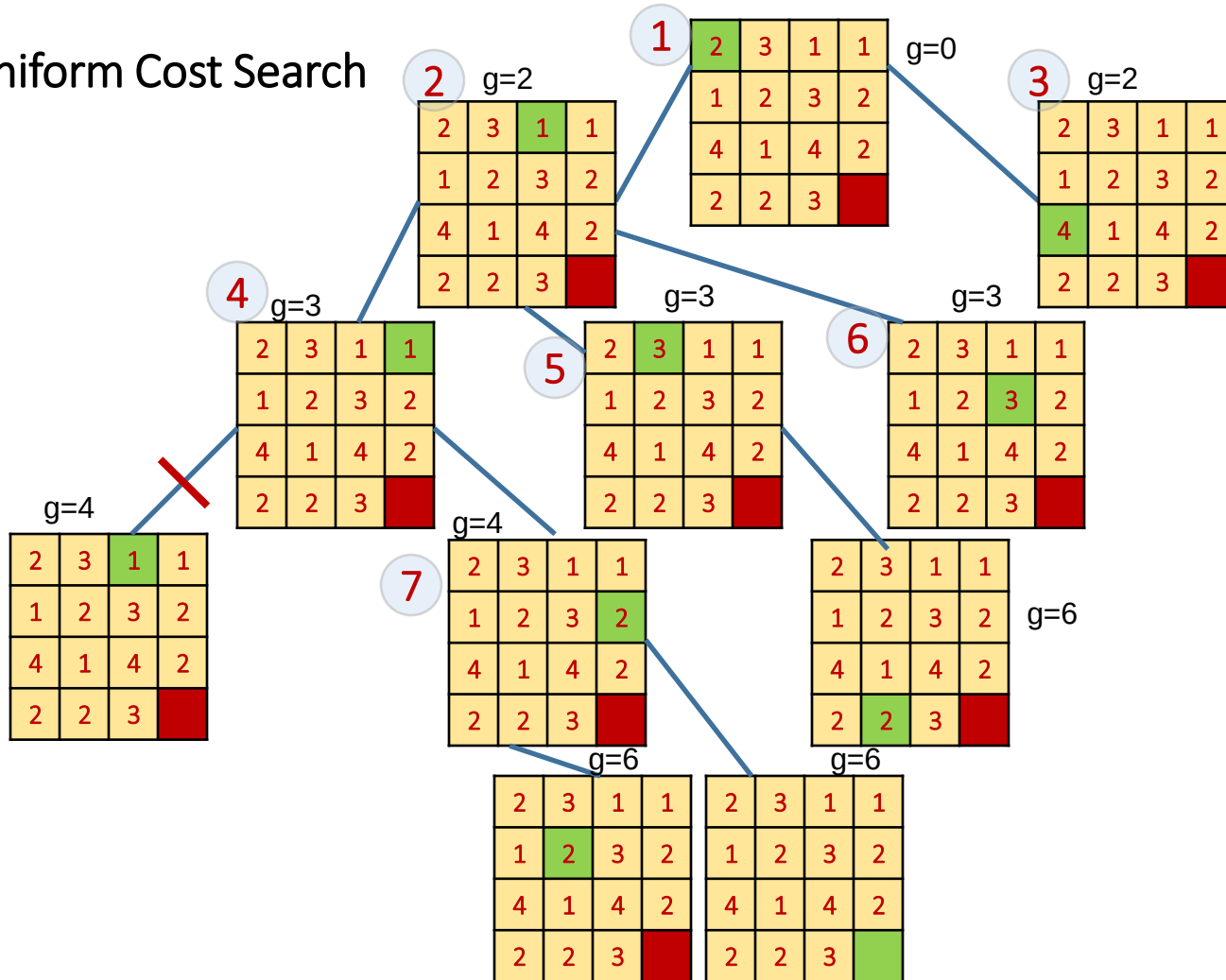


FRONTIER

[(1,1)₀]
[(1,3)₂, (3,1)₂]
[(3,1)₂, (1,4)₃, (1,2)₃, (2,3)₃]
[(1,4)₃, (1,2)₃, (2,3)₃]
[(1,2)₃, (2,3)₃, (2,4)₄]
[(2,3)₃, (2,4)₄, (4,2)₆]
[(2,4)₄, (4,2)₆]

EXPLORED SET: (1,1), (1,3), (3,1), (1,4), (1,2), (2,3), (2,4)

Ex 1: Uniform Cost Search



FRONTIER

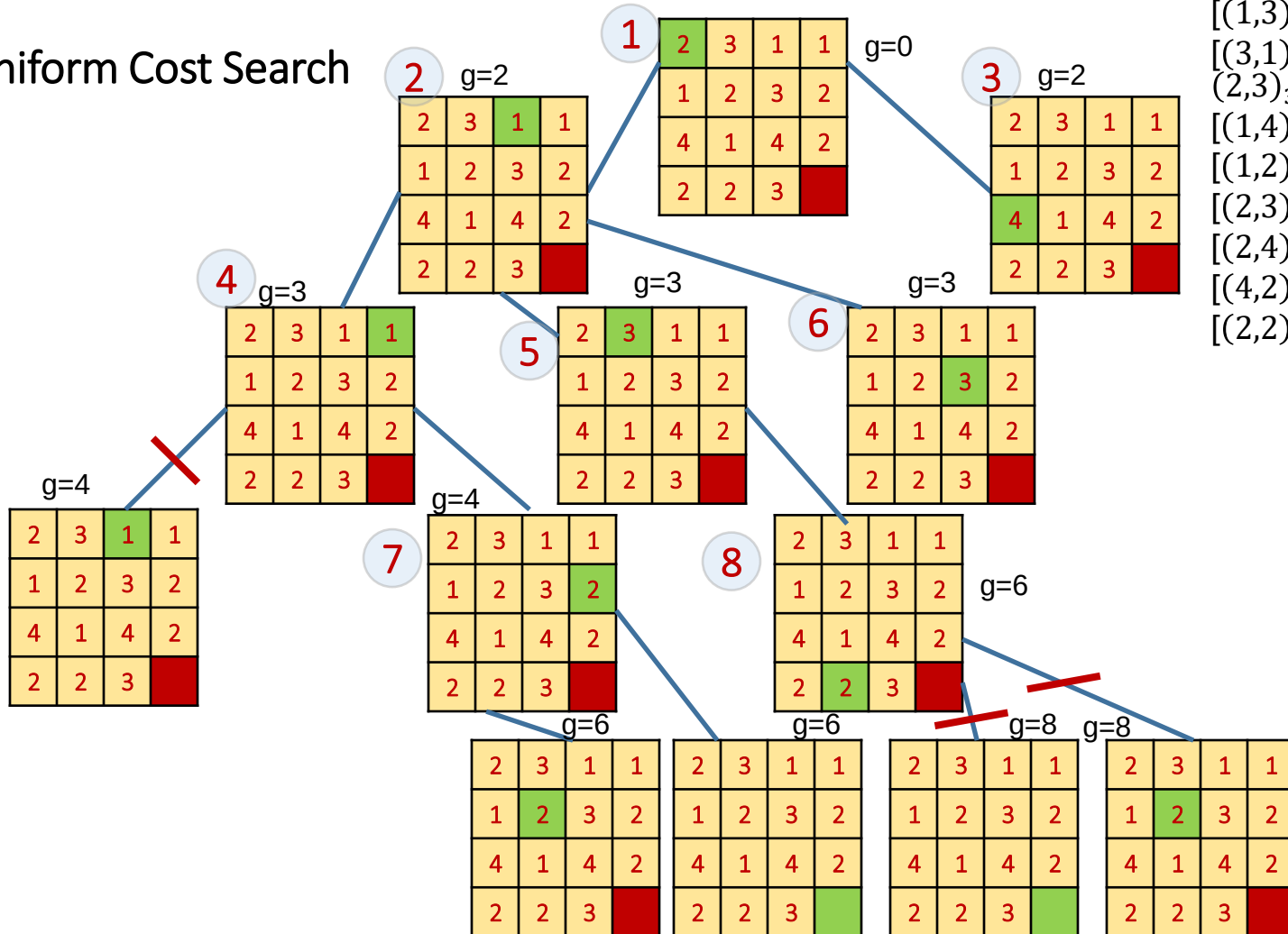
[(1,1)₀]
 [(1,3)₂, (3,1)₂]
 [(3,1)₂, (1,4)₃, (1,2)₃, (2,3)₃]
 [(1,4)₃, (1,2)₃, (2,3)₃]
 [(1,2)₃, (2,3)₃, (2,4)₄]
 [(2,3)₃, (2,4)₄, (4,2)₆]
 [(2,4)₄, (4,2)₆]
 [(4,2)₆, (2,2)₆, (4,4)₆]

EXPLORED SET: (1,1), (1,3), (3,1), (1,4), (1,2), (2,3), (2,4), (4,2)

FRONTIER

[(1,1)₀]
 [(1,3)₂, (3,1)₂]
 [(3,1)₂, (1,4)₃, (1,2)₃,
 (2,3)₃]
 [(1,4)₃, (1,2)₃, (2,3)₃]
 [(1,2)₃, (2,3)₃, (2,4)₄]
 [(2,3)₃, (2,4)₄, (4,2)₆]
 [(2,4)₄, (4,2)₆]
 [(4,2)₆, (2,2)₆, (4,4)₆]
 [(2,2)₆, (4,4)₆]

Ex 1: Uniform Cost Search

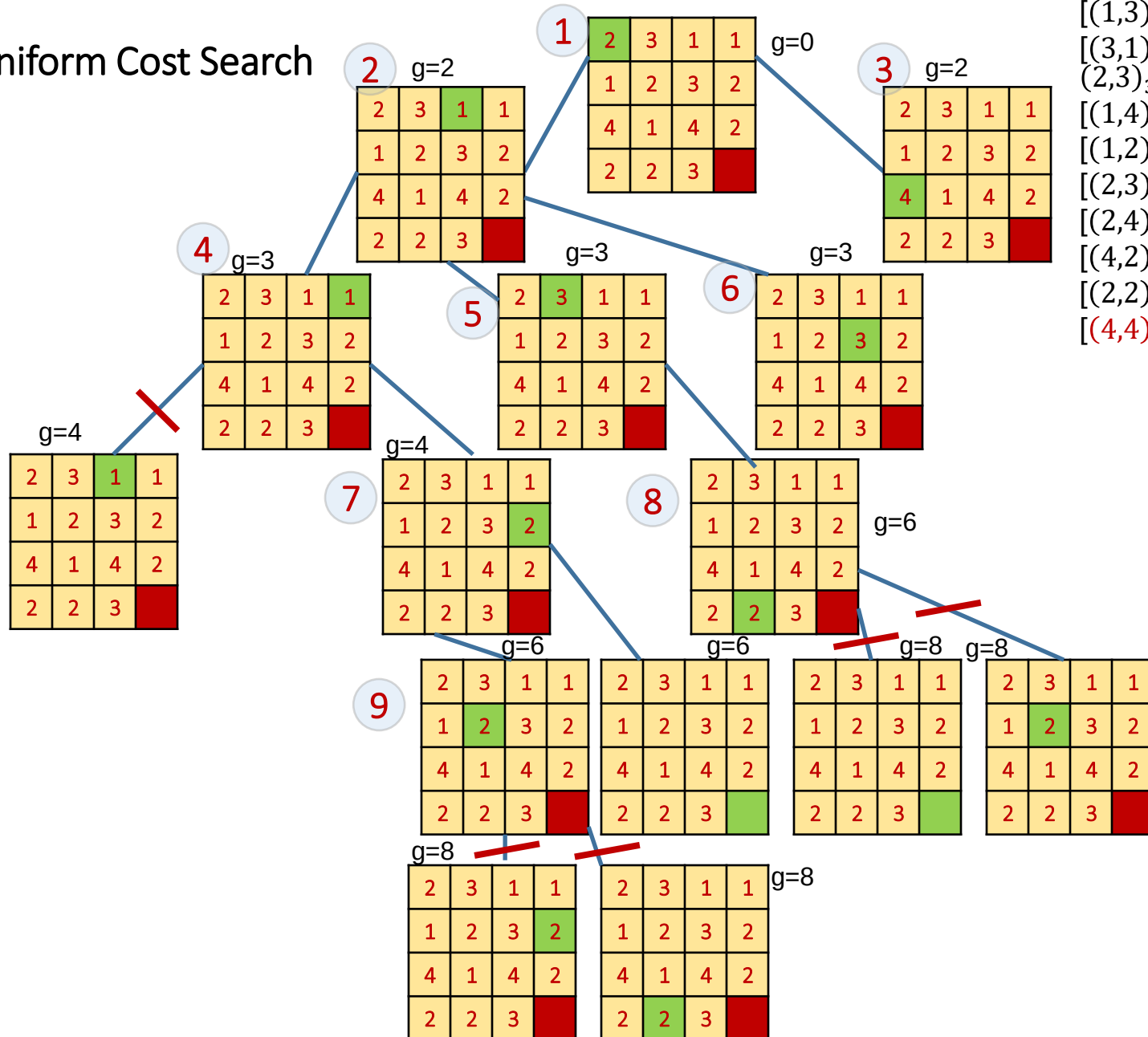


EXPLORED SET: (1,1), (1,3), (3,1), (1,4), (1,2), (2,3), (2,4), (4,2), (2,2)

FRONTIER

[(1,1)₀]
 [(1,3)₂, (3,1)₂]
 [(3,1)₂, (1,4)₃, (1,2)₃, (2,3)₃]
 [(1,4)₃, (1,2)₃, (2,3)₃]
 [(1,2)₃, (2,3)₃, (2,4)₄]
 [(2,3)₃, (2,4)₄, (4,2)₆]
 [(2,4)₄, (4,2)₆]
 [(4,2)₆, (2,2)₆, (4,4)₆]
 [(2,2)₆, (4,4)₆]
 [(4,4)₆]

Ex 1: Uniform Cost Search



FRONTIER

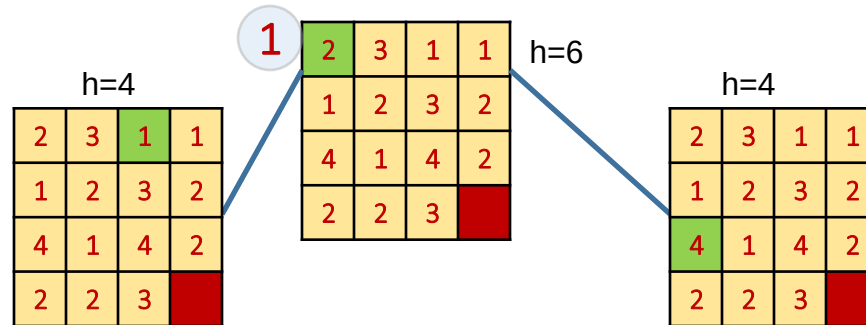
$[(1,1)_6]$

Ex 1: Greedy Search

2	3	1	1
1	2	3	2
4	1	4	2
2	2	3	

h=6

Ex 1: Greedy Search

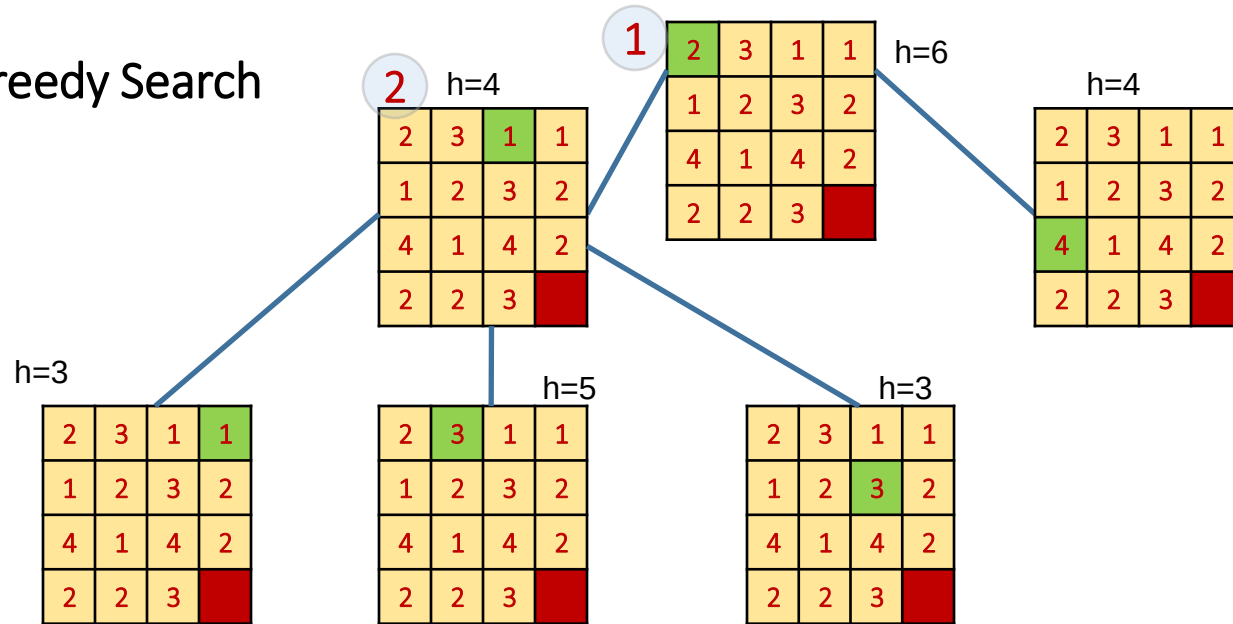


FRONTIER

$[(1,1)_6]$

$[(1,3)_4, (3,1)_4]$

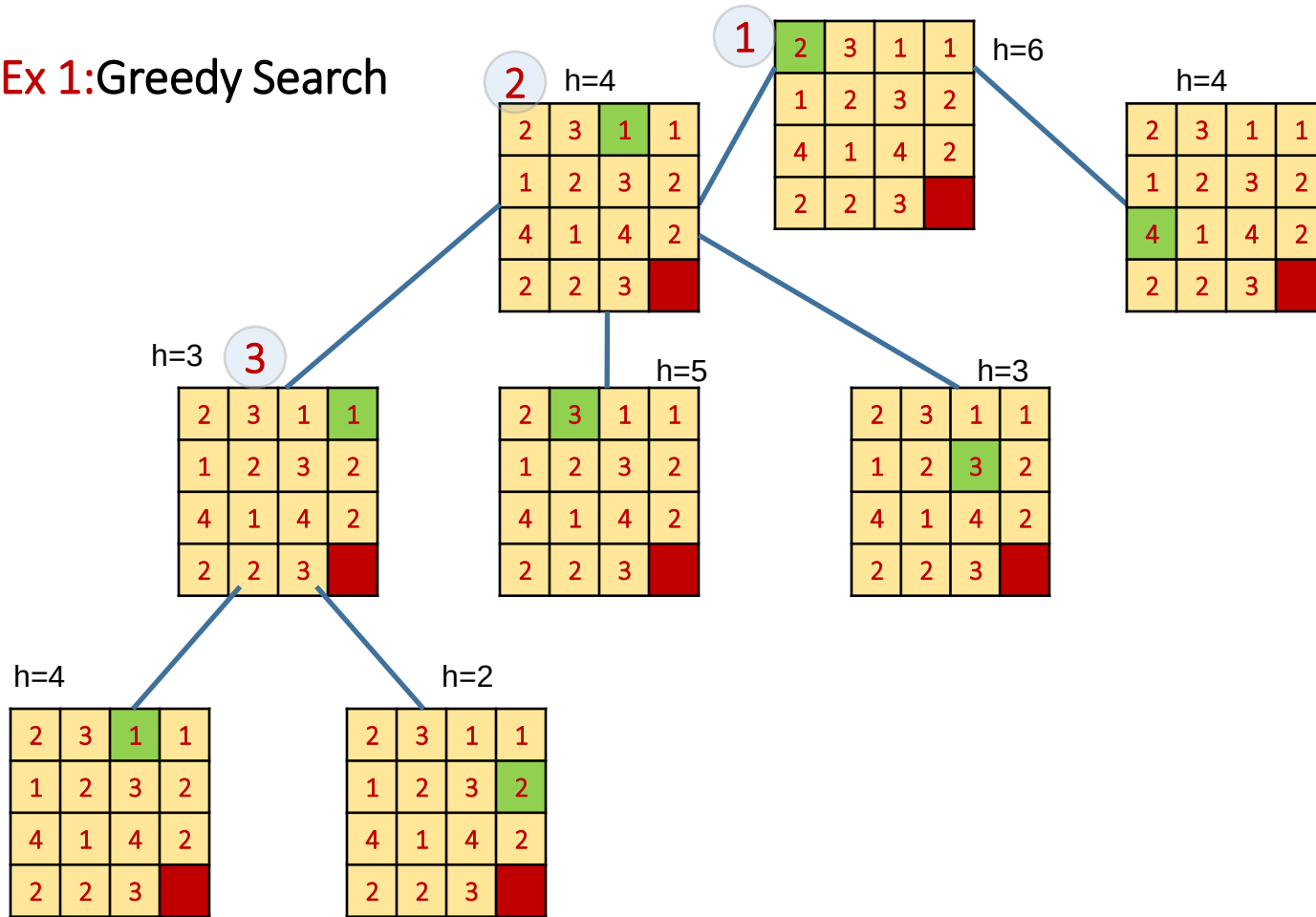
Ex 1: Greedy Search



FRONTIER

$[(1,1)_6]$
 $[(1,3)_4, (3,1)_4]$
 $[(1,4)_3, (2,3)_3, (3,1)_4, (1,2)_5]$

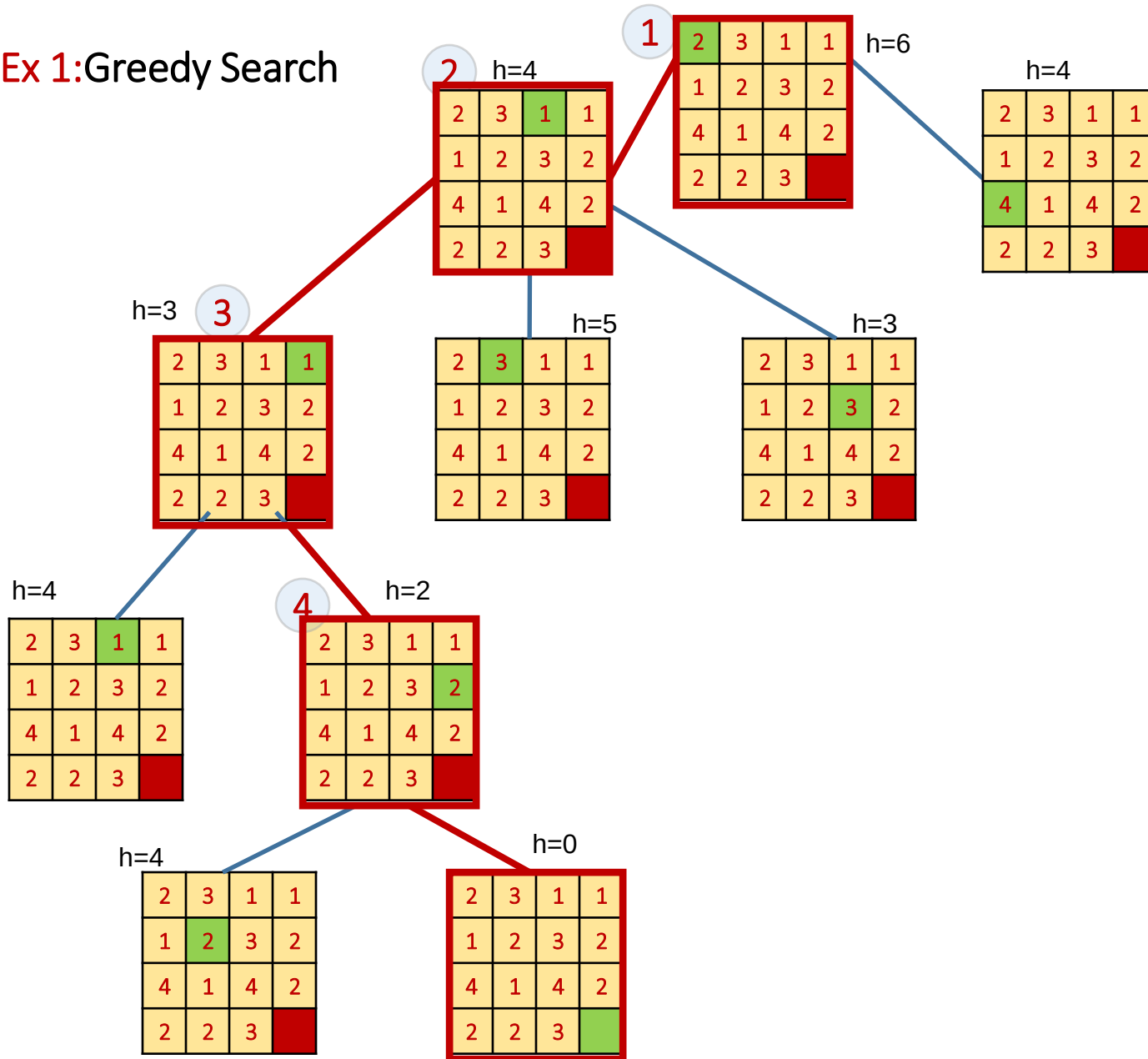
Ex 1: Greedy Search



FRONTIER

$[(1,1)_6]$
 $[(1,3)_4, (3,1)_4]$
 $[(1,4)_3, (2,3)_3, (3,1)_4, (1,2)_5]$
 $[(2,4)_2, (2,3)_3, (3,1)_4, (1,3)_4, (1,2)_5]$

Ex 1: Greedy Search



FRONTIER

$[(1,1)_6]$
 $[(1,3)_4, (3,1)_4]$
 $[(1,4)_3, (2,3)_3, (3,1)_4, (1,2)_5]$
 $[(2,4)_2, (2,3)_3, (3,1)_4, (1,3)_4, (1,2)_5]$
 $[(2,3)_3, (3,1)_4, (1,3)_4, (2,2)_4, (1,2)_5]$

EXPLORED SET:

FRONTIER

$[(1,1)_6]$

Ex 1: Greedy Search

2	3	1	1
1	2	3	2
4	1	4	2
2	2	3	

h=6

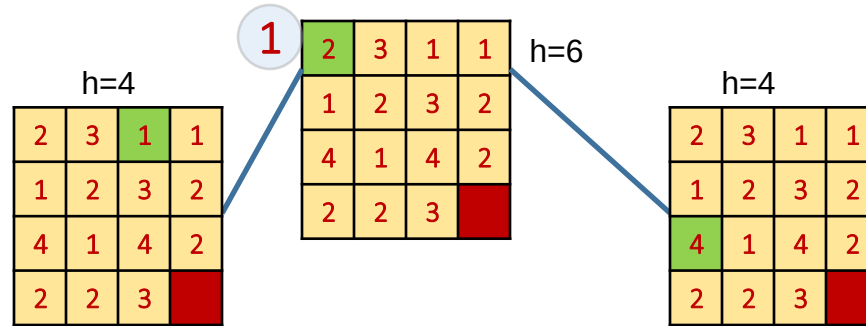
EXPLORED SET: (1,1)

FRONTIER

$[(1,1)_6]$

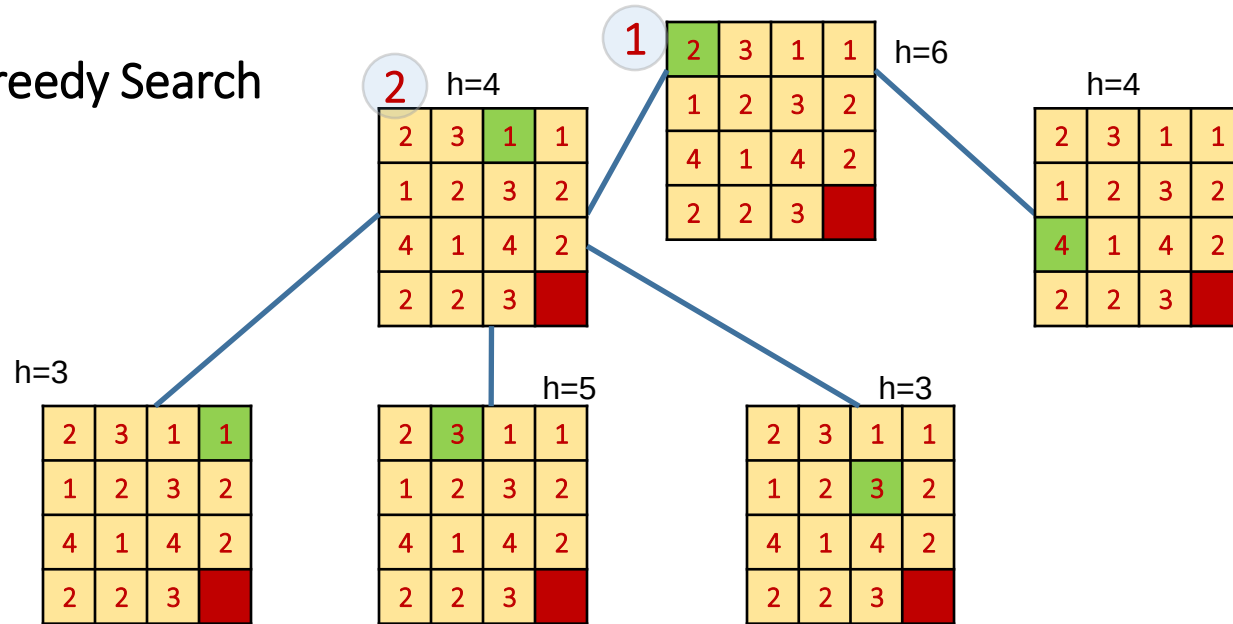
$[(1,3)_4, (3,1)_4]$

Ex 1: Greedy Search



EXPLORED SET: (1,1), (1,3)

Ex 1: Greedy Search



FRONTIER

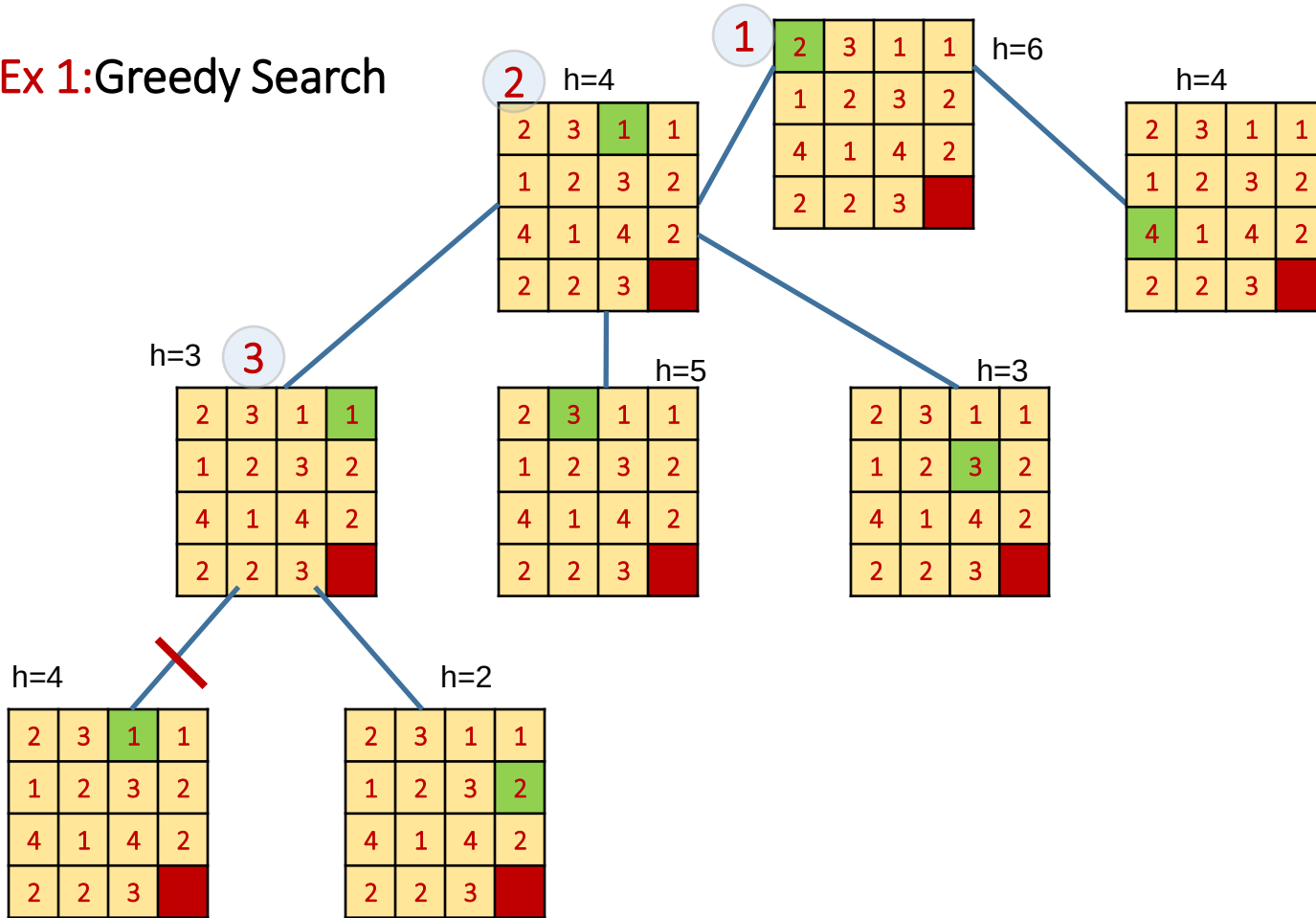
$[(1,1)_6]$
 $[(1,3)_4, (3,1)_4]$
 $[(1,4)_3, (2,3)_3, (3,1)_4, (1,2)_5]$

EXPLORED SET: (1,1), (1,3), (1,4)

FRONTIER

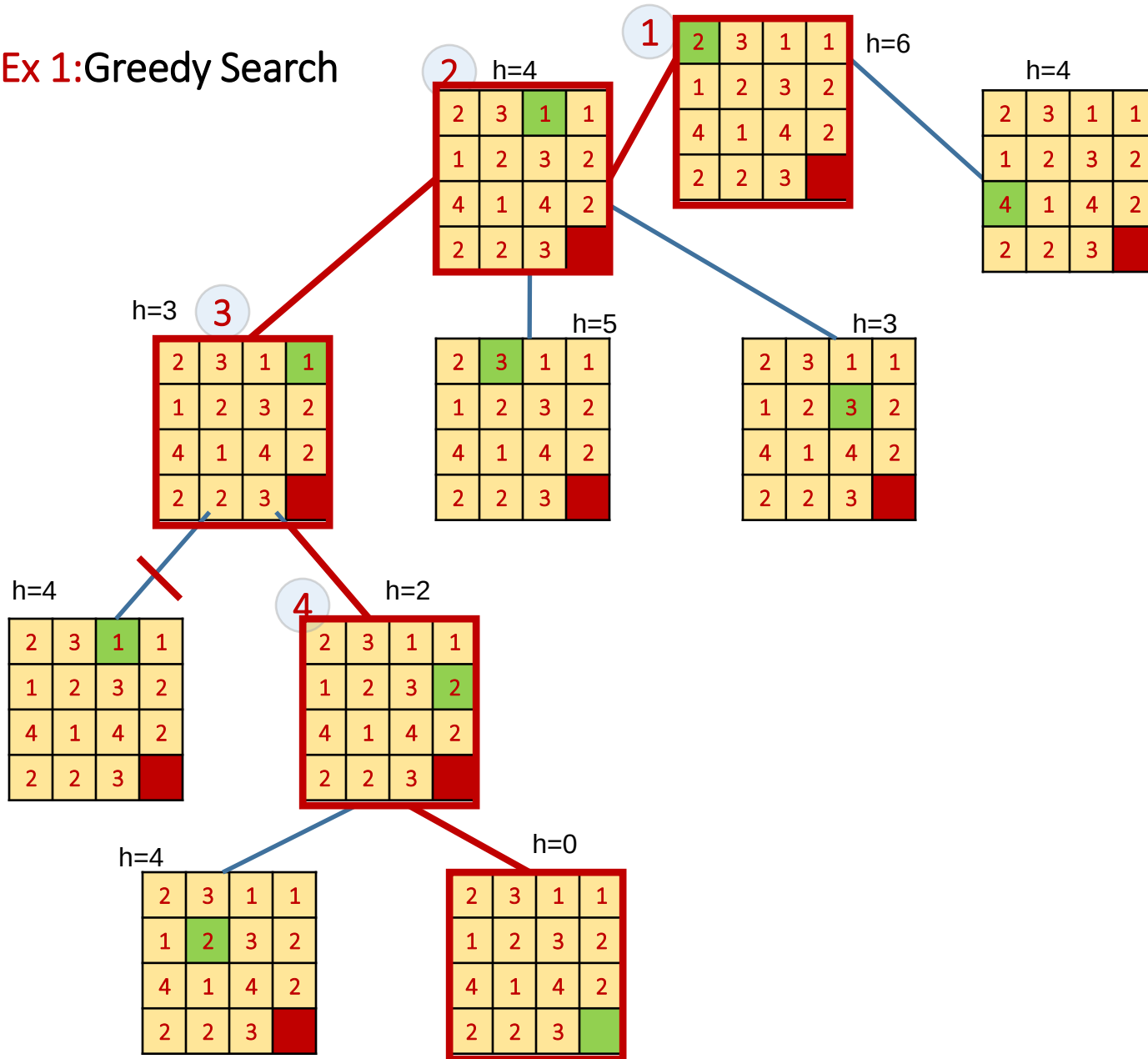
$[(1,1)_6]$
 $[(1,3)_4, (3,1)_4]$
 $[(1,4)_3, (2,3)_3, (3,1)_4, (1,2)_5]$
 $[(2,4)_2, (2,3)_3, (3,1)_4, (1,2)_5]$

Ex 1: Greedy Search



EXPLORED SET: (1,1), (1,3), (1,4)

Ex 1: Greedy Search



FRONTIER

[(1,1)₆]
 [(1,3)₄, (3,1)₄]
 [(1,4)₃, (2,3)₃, (3,1)₄,
 (1,2)₅]
 [(2,4)₂, (2,3)₃, (3,1)₄,
 (1,2)₅]
 [(2,3)₃, (3,1)₄,
 (2,2)₄, (1,2)₅]

FRONTIER

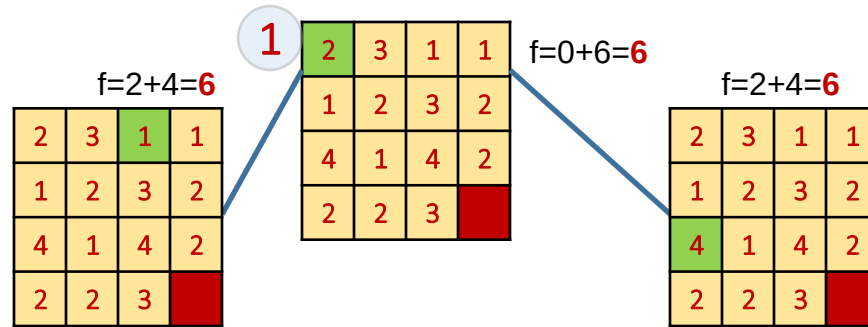
$[(1,1)_6]$

Ex 1: A* Search

2	3	1	1
1	2	3	2
4	1	4	2
2	2	3	

$$f=0+6=6$$

Ex 1: A* Search

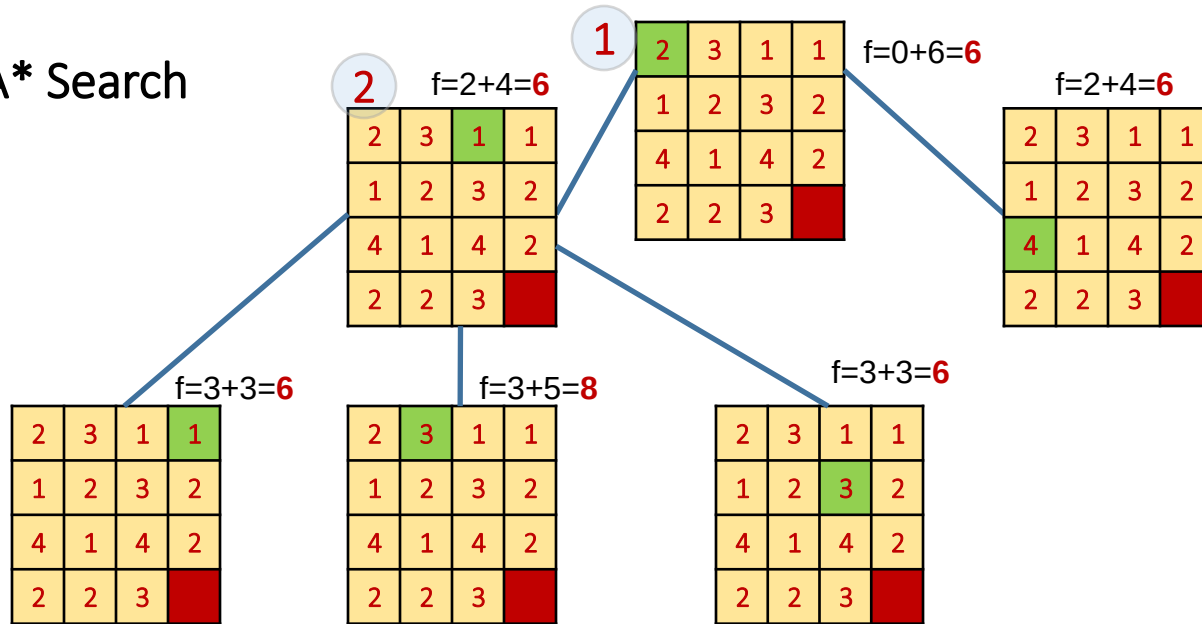


FRONTIER

$[(1,1)_6]$

$[(1,3)_6, (3,1)_6]$

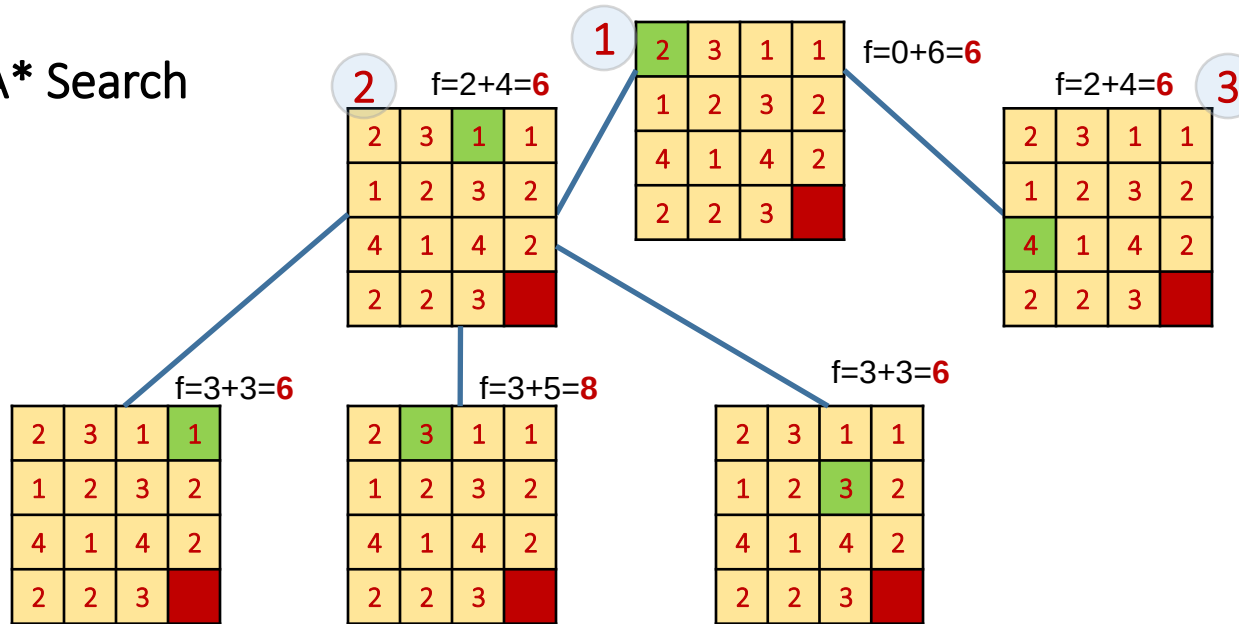
Ex 1: A* Search



FRONTIER

$[(1,1)_6]$
 $[(1,3)_6, (3,1)_6]$
 $[(3,1)_6, (1,4)_6, (2,3)_6, (1,2)_8]$

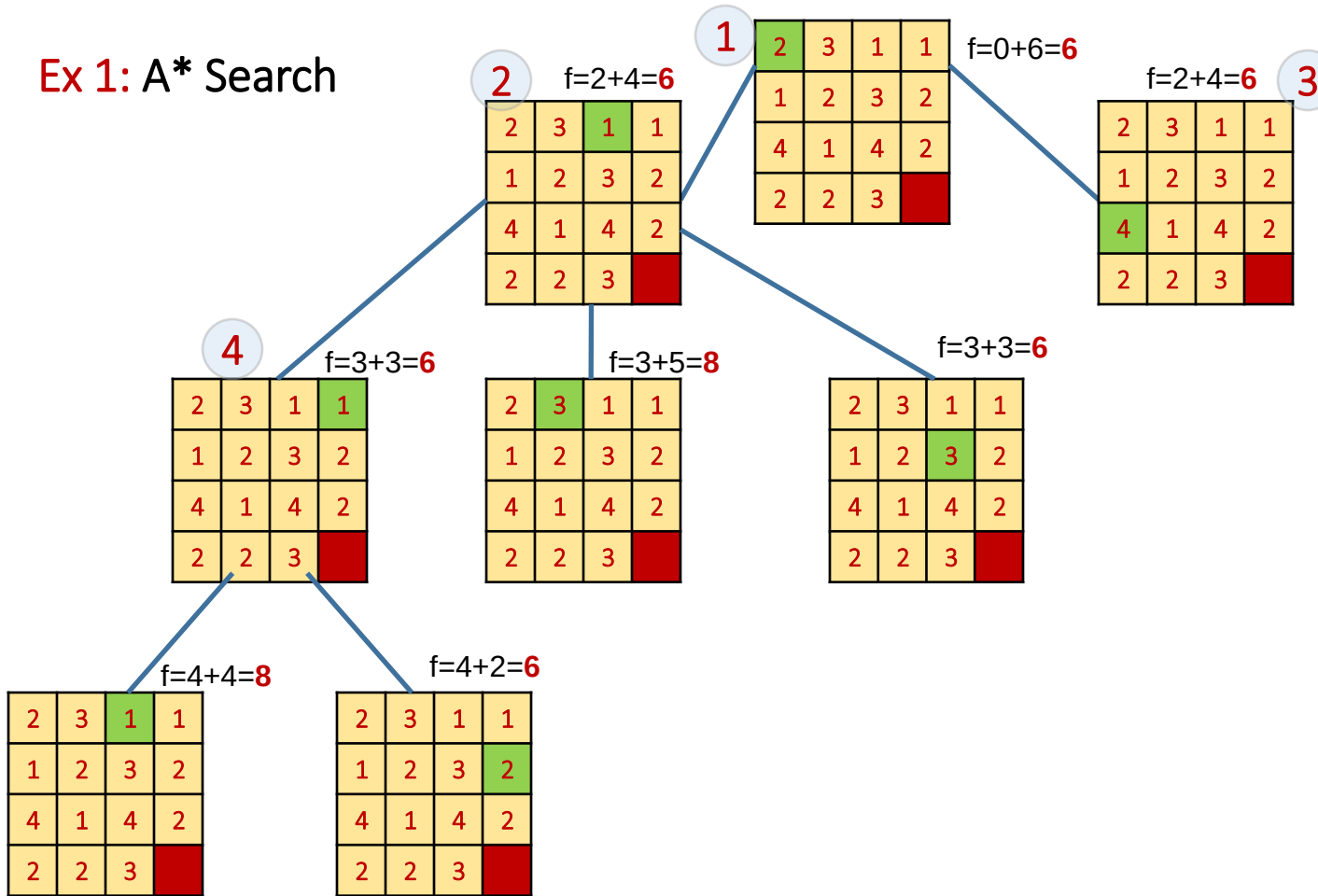
Ex 1: A* Search



FRONTIER

$[(1,1)_6]$
 $[(1,3)_6, (3,1)_6]$
 $[(3,1)_6, (1,4)_6, (2,3)_6, (1,2)_8]$
 $[(1,4)_6, (2,3)_6, (1,2)_8]$

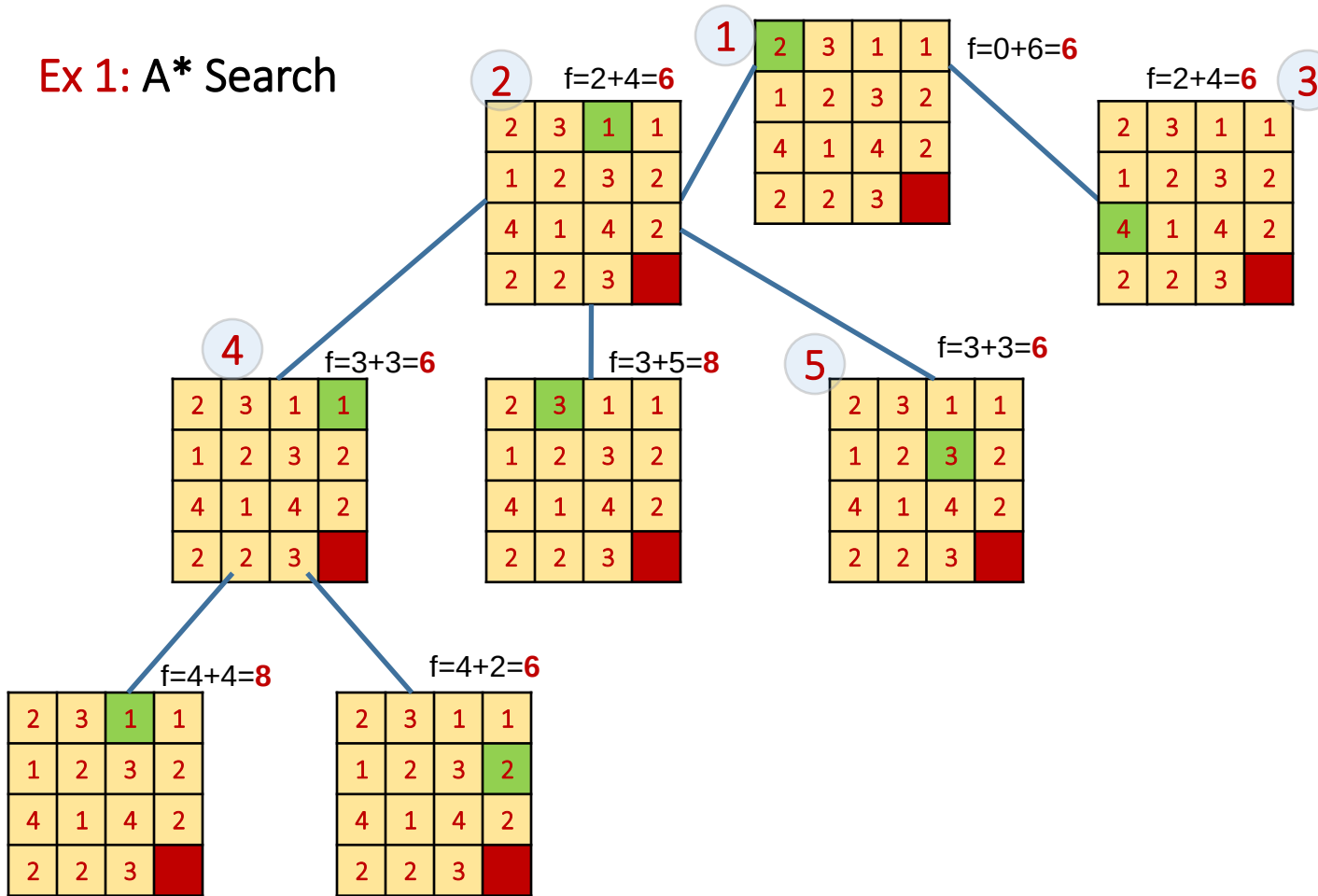
Ex 1: A* Search



FRONTIER

$[(1,1)_6]$
 $[(1,3)_6, (3,1)_6]$
 $[(3,1)_6, (1,4)_6, (2,3)_6, (1,2)_8]$
 $[(1,4)_6, (2,3)_6, (1,2)_8]$
 $[(2,3)_6, (2,4)_6, (1,2)_8, (1,3)_8]$

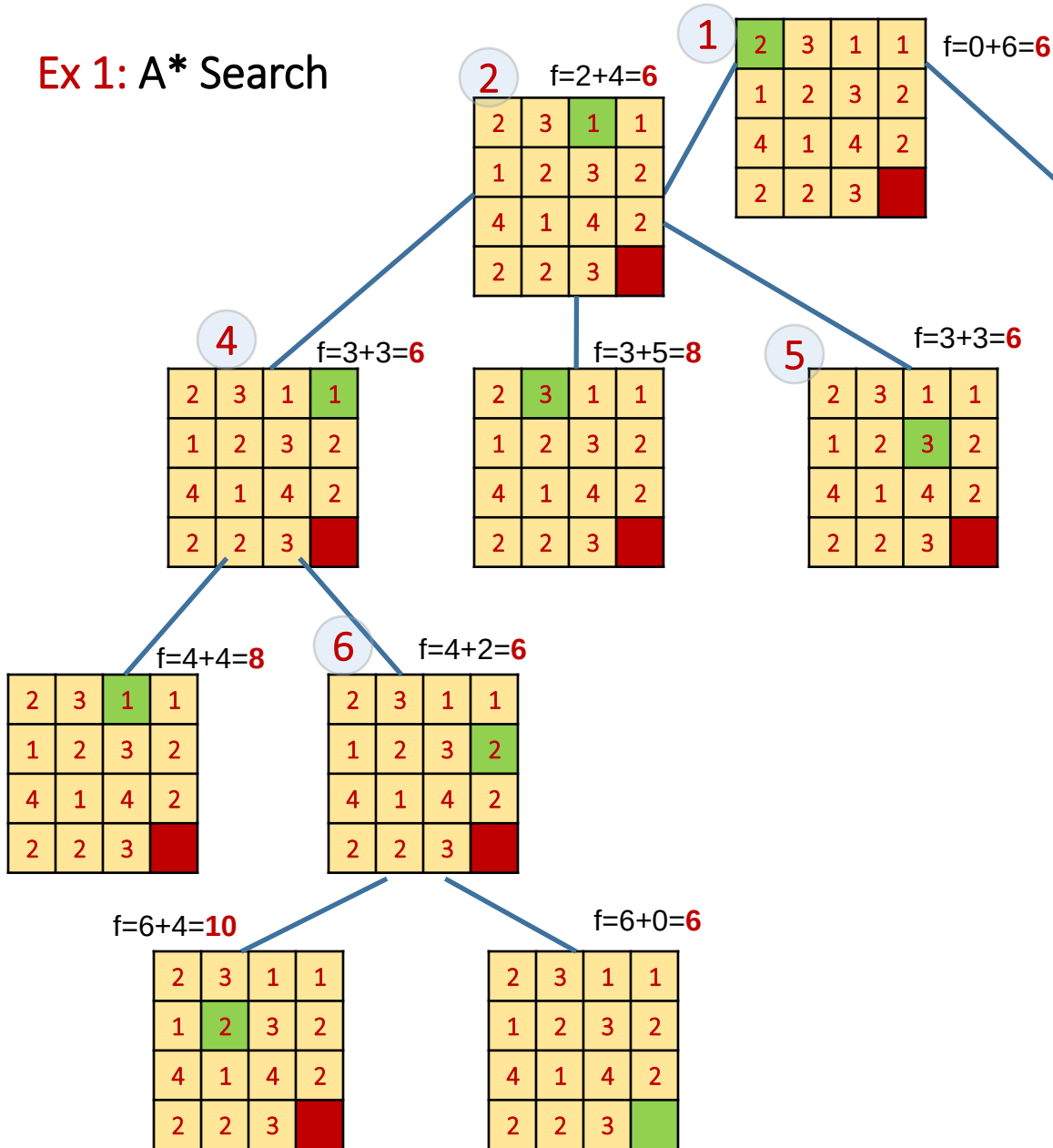
Ex 1: A* Search



FRONTIER

$[(1,1)_6]$
 $[(1,3)_6, (3,1)_6]$
 $[(3,1)_6, (1,4)_6, (2,3)_6, (1,2)_8]$
 $[(1,4)_6, (2,3)_6, (1,2)_8]$
 $[(2,3)_6, (2,4)_6, (1,2)_8, (1,3)_8]$
 $[(2,4)_6, (1,2)_8, (1,3)_8]$

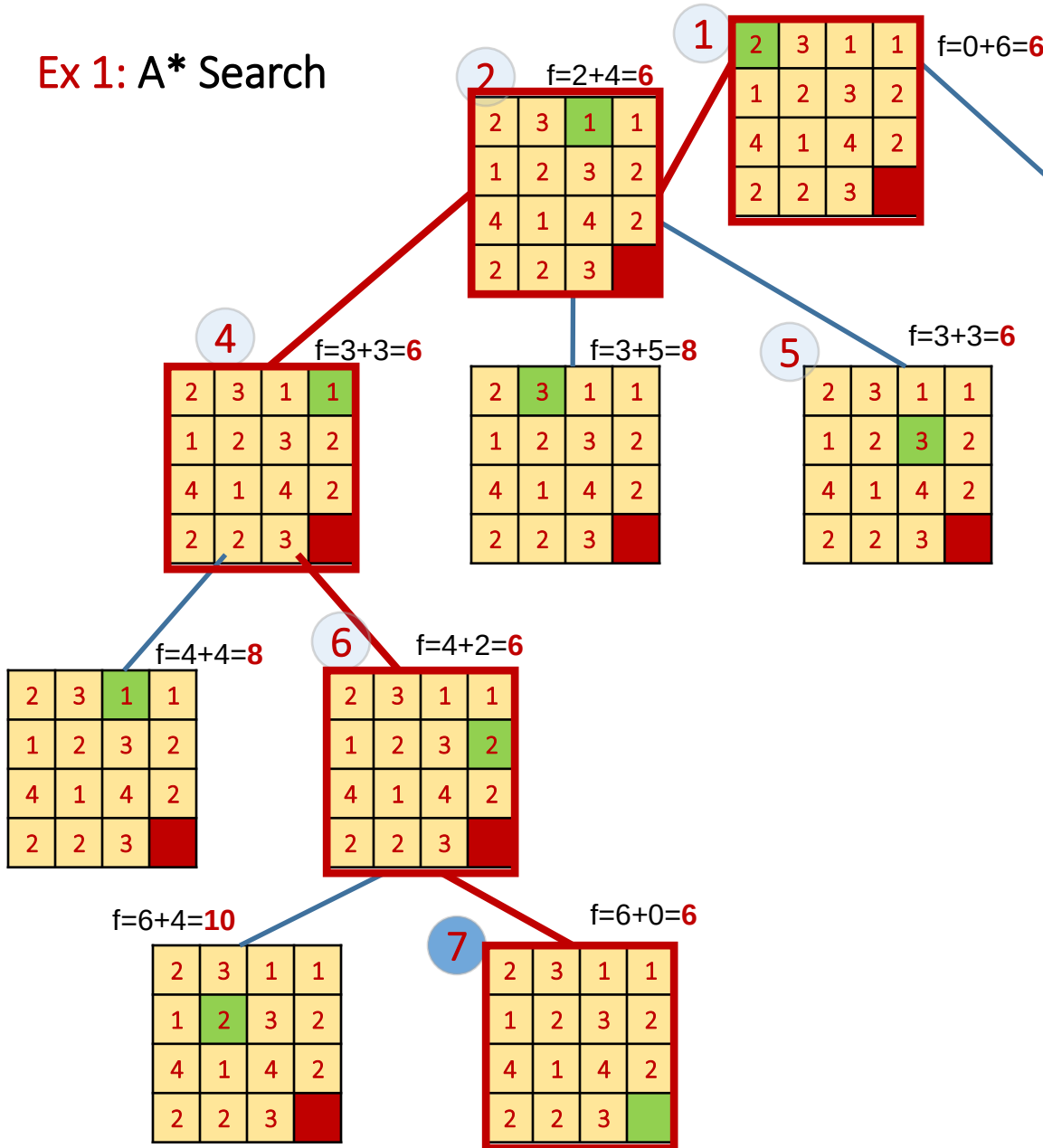
Ex 1: A* Search



FRONTIER

$[(1,1)_6]$
 $[(1,3)_6, (3,1)_6]$
 $[(3,1)_6, (1,4)_6, (2,3)_6, (1,2)_8]$
 $[(1,4)_6, (2,3)_6, (1,2)_8]$
 $[(2,3)_6, (2,4)_6, (1,2)_8, (1,3)_8]$
 $[(2,4)_6, (1,2)_8, (1,3)_8]$
 $[(4,4)_6, (1,2)_8, (1,3)_8, (2,2)_{10}]$

Ex 1: A* Search



FRONTIER

$[(1,1)_6]$
 $[(1,3)_6, (3,1)_6]$
 $[(3,1)_6, (1,4)_6, (2,3)_6, (1,2)_8]$
 $[(1,4)_6, (2,3)_6, (1,2)_8]$
 $[(2,3)_6, (2,4)_6, (1,2)_8, (1,3)_8]$
 $[(2,4)_6, (1,2)_8, (1,3)_8]$
 $[(4,4)_6, (1,2)_8, (1,3)_8, (2,2)_{10}]$
 $[(1,2)_8, (1,3)_8, (2,2)_{10}]$

EXPLORED SET:

FRONTIER

$[(1,1)_6]$

Ex 1: A* Search

2	3	1	1
1	2	3	2
4	1	4	2
2	2	3	

$$f=0+6=6$$

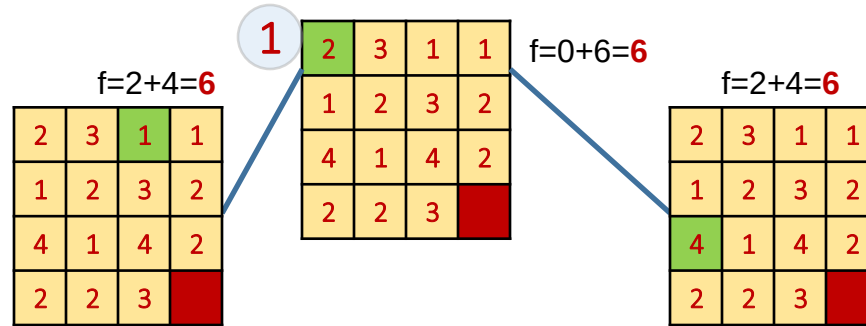
EXPLORED SET: (1,1)

FRONTIER

$[(1,1)_6]$

$[(1,3)_6, (3,1)_6]$

Ex 1: A* Search

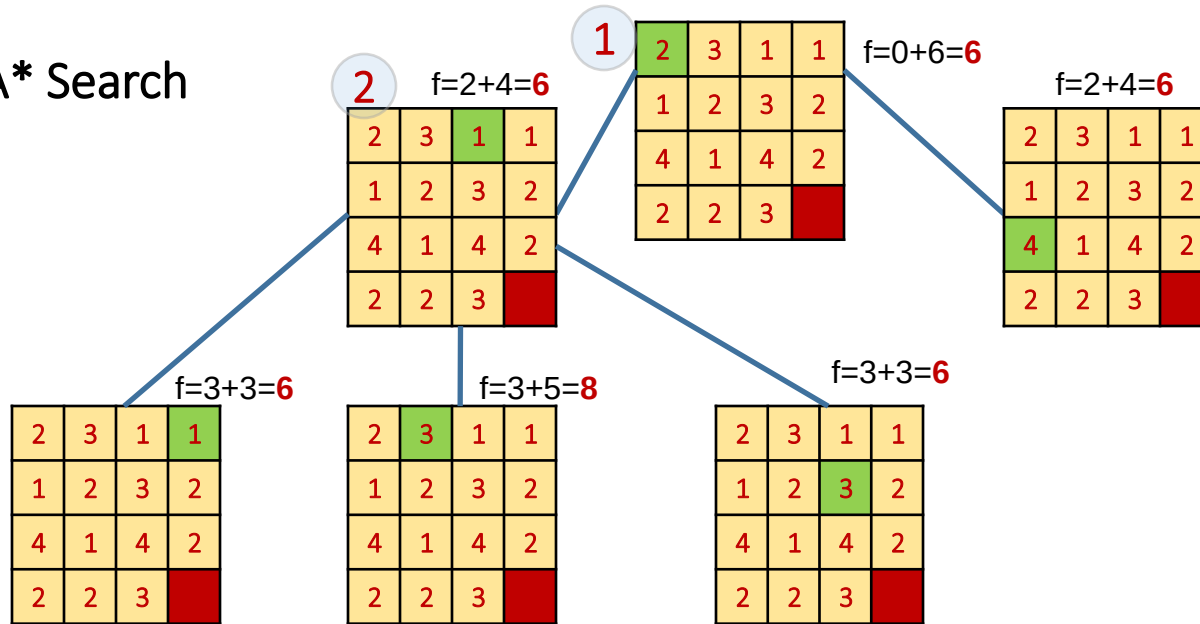


EXPLORED SET: (1,1), (1,3)

FRONTIER

$[(1,1)_6]$
 $[(1,3)_6, (3,1)_6]$
 $[(3,1)_6, (1,4)_6, (2,3)_6, (1,2)_8]$

Ex 1: A* Search

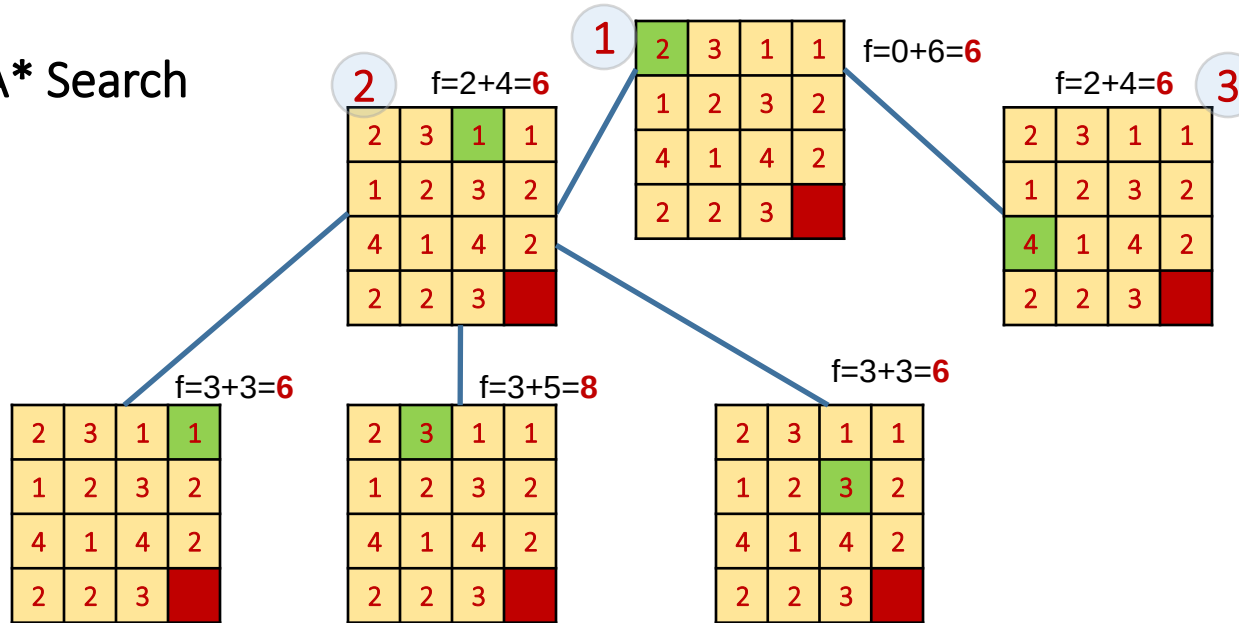


EXPLORED SET: (1,1), (1,3), (3,1)

FRONTIER

$[(1,1)_6]$
 $[(1,3)_6, (3,1)_6]$
 $[(3,1)_6, (1,4)_6, (2,3)_6, (1,2)_8]$
 $[(1,4)_6, (2,3)_6, (1,2)_8]$

Ex 1: A* Search

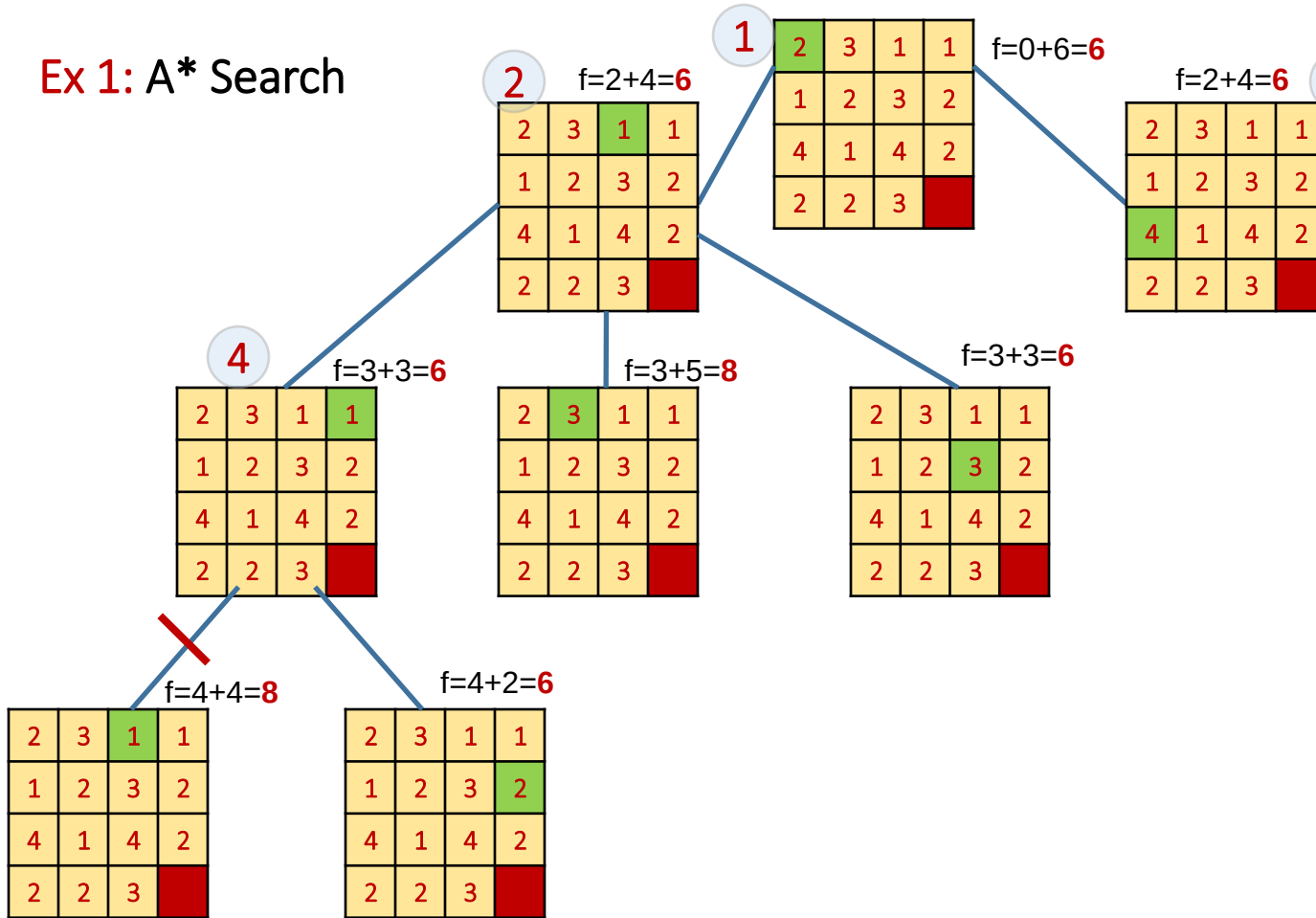


EXPLORED SET: (1,1), (1,3), (3,1), (1,4)

FRONTIER

$[(1,1)_6]$
 $[(1,3)_6, (3,1)_6]$
 $[(3,1)_6, (1,4)_6, (2,3)_6, (1,2)_8]$
 $[(1,4)_6, (2,3)_6, (1,2)_8]$
 $[(2,3)_6, (2,4)_6, (1,2)_8]$

Ex 1: A* Search

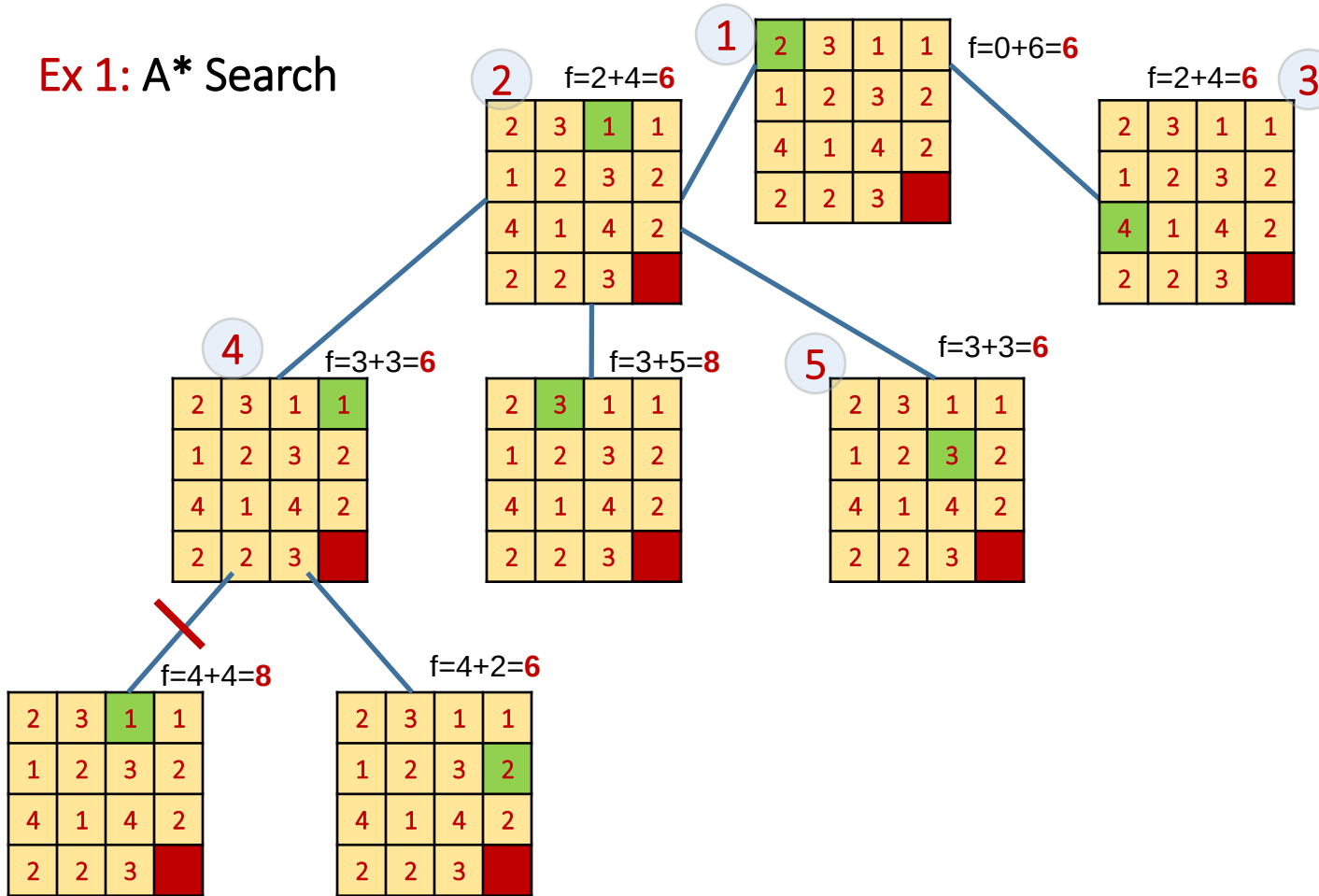


EXPLORED SET: (1,1), (1,3), (3,1), (1,4), (2,3)

FRONTIER

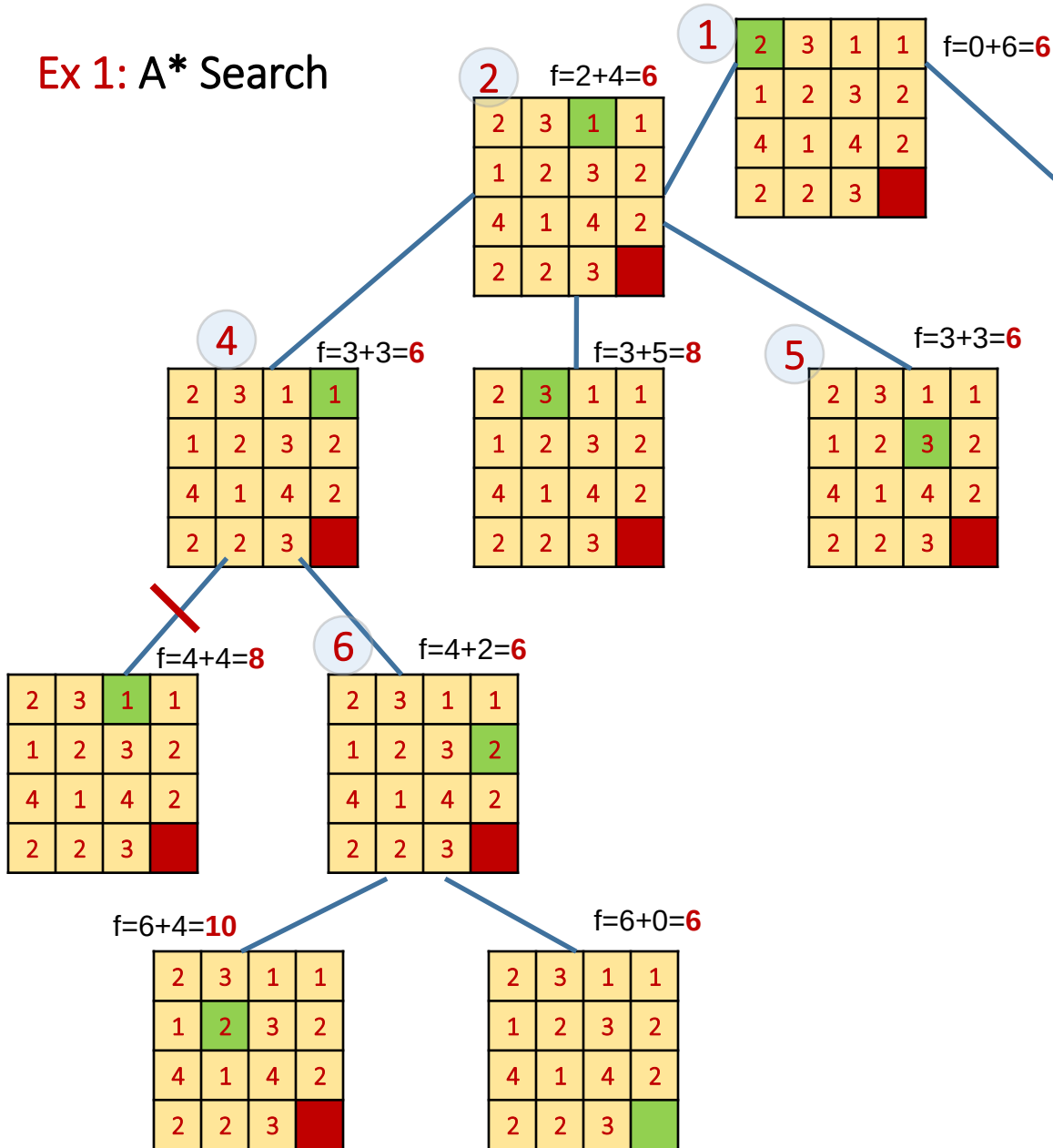
$[(1,1)_6]$
 $[(1,3)_6, (3,1)_6]$
 $[(3,1)_6, (1,4)_6, (2,3)_6, (1,2)_8]$
 $[(1,4)_6, (2,3)_6, (1,2)_8]$
 $[(2,3)_6, (2,4)_6, (1,2)_8,]$
 $[(2,4)_6, (1,2)_8,]$

Ex 1: A* Search



EXPLORED SET: (1,1), (1,3), (3,1), (1,4), (2,3), (2,4)

Ex 1: A* Search

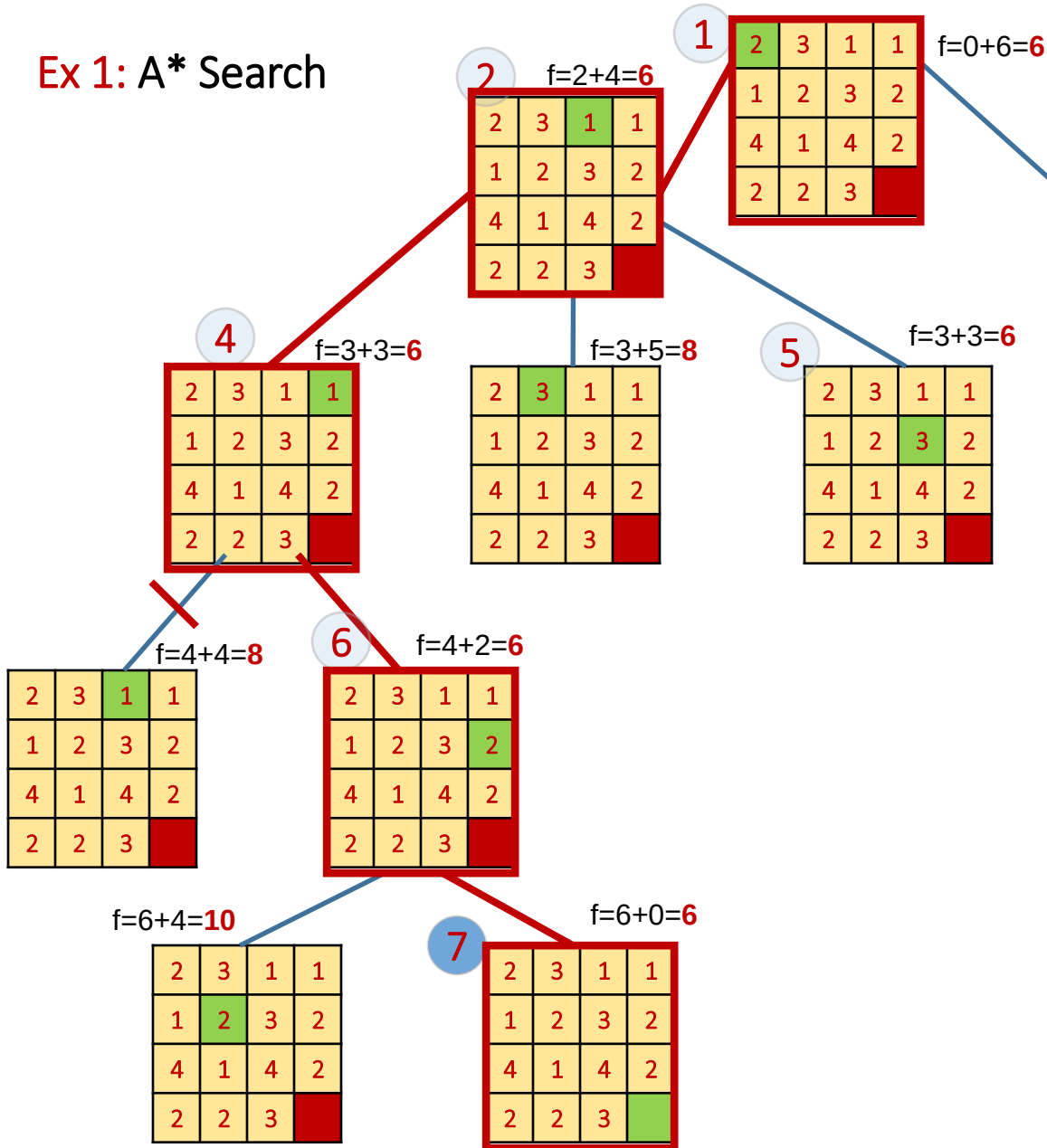


FRONTIER

[(1,1)₆]
 [(1,3)₆, (3,1)₆]
 [(3,1)₆, (1,4)₆, (2,3)₆,
 (1,2)₈]
 [(1,4)₆, (2,3)₆, (1,2)₈]
 [(2,3)₆, (2,4)₆, (1,2)₈,]
 [(2,4)₆, (1,2)₈,]
 [(4,4)₆, (1,2)₈,
 (2,2)₁₀]

EXPLORED SET: (1,1), (1,3), (3,1), (1,4), (2,3), (2,4)

Ex 1: A* Search



FRONTIER

[(1,1)₆]
 [(1,3)₆, (3,1)₆]
 [(3,1)₆, (1,4)₆, (2,3)₆, (1,2)₈]
 [(1,4)₆, (2,3)₆, (1,2)₈]
 [(2,3)₆, (2,4)₆, (1,2)₈,]
 [(2,4)₆, (1,2)₈,]
 [(4,4)₆, (1,2)₈,
 (2,2)₁₀]
 [(1,2)₈, (2,2)₁₀]